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DATA VISUALIZATION PRACTICAL - 10

PRACTICAL – 10

AIM: Data Preparation and Transformation using Tableau Prep
a) Connect and load data into Tableau Prep
b) Clean and profile data
c) Group and standardize values
d) Merge mismatched fields
e) Perform aggregations in Prep
f) Apply filters and transformations
g) Create and automate data preparation flows

A: Connect and Load Data into Tableau Prep

Objective: Establish connections to the local CSV data sources (Student_Assessments_Term1.csv, Student_Assessments_Term2.csv, and Course_Credit_Master.csv) and initiate a visual data preparation workflow.

Tableau Prep Builder utilizes an input step pipeline architecture. Connecting multiple disparate files into a unified workspace establishes the foundation for schema validation, data type detection, and downstream data profiling.

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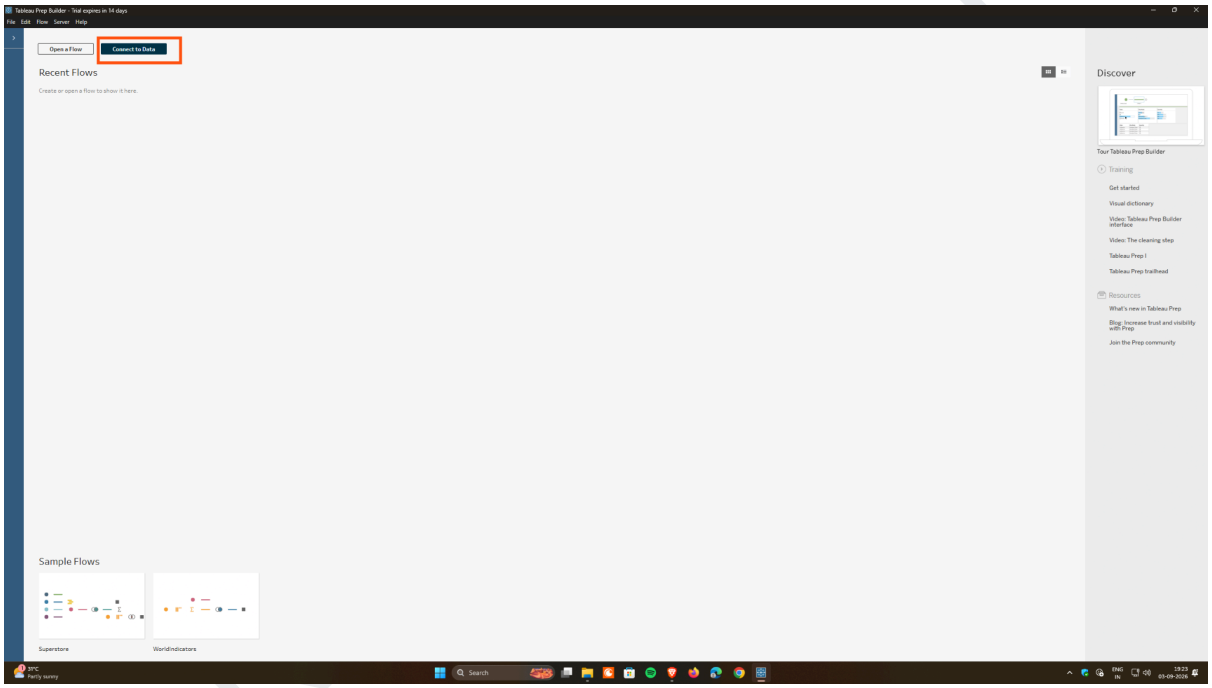
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Step 1: Launch **Tableau Prep Builder** and locate the **Connections** pane on the left side of the start screen.

Step 2: In the **Connections** pane, click the **+ (Add Connection)** button and select **Student_Assessments_Term1.csv**.

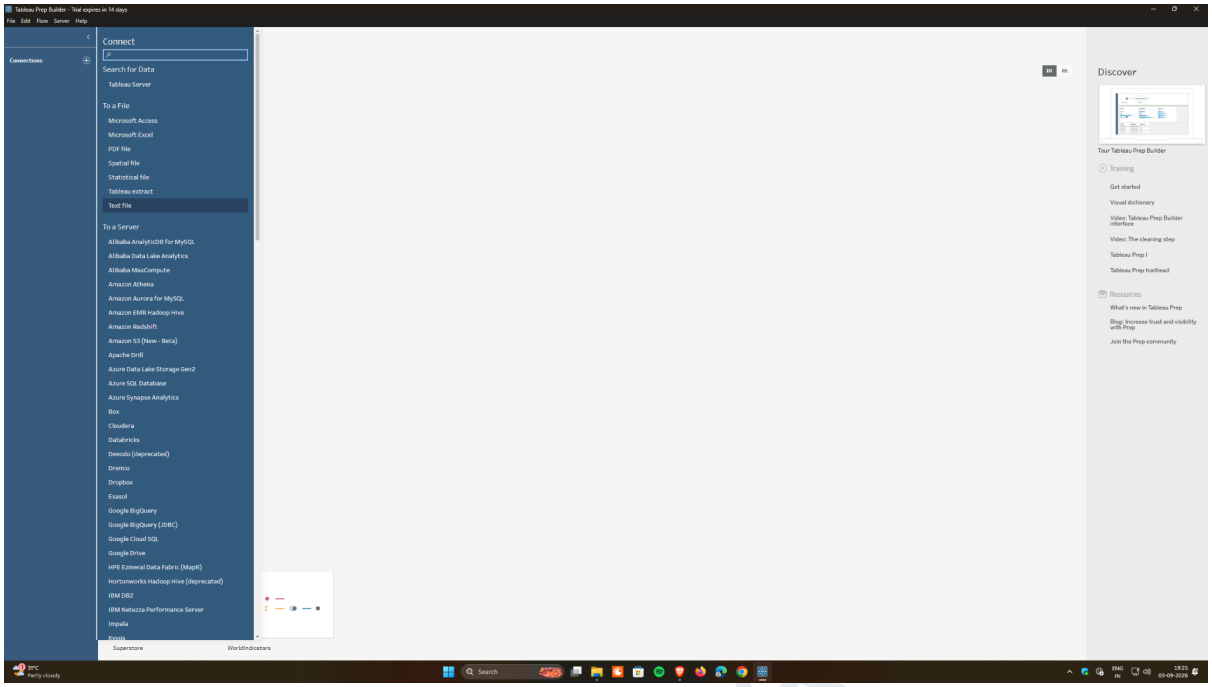


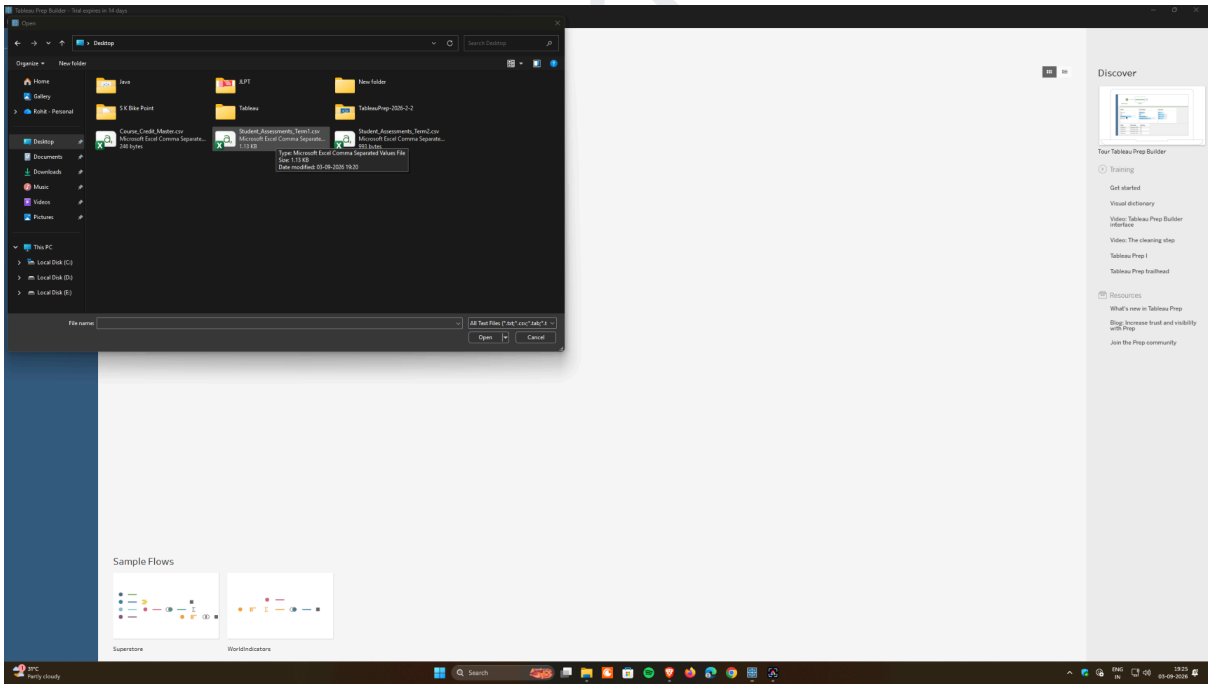
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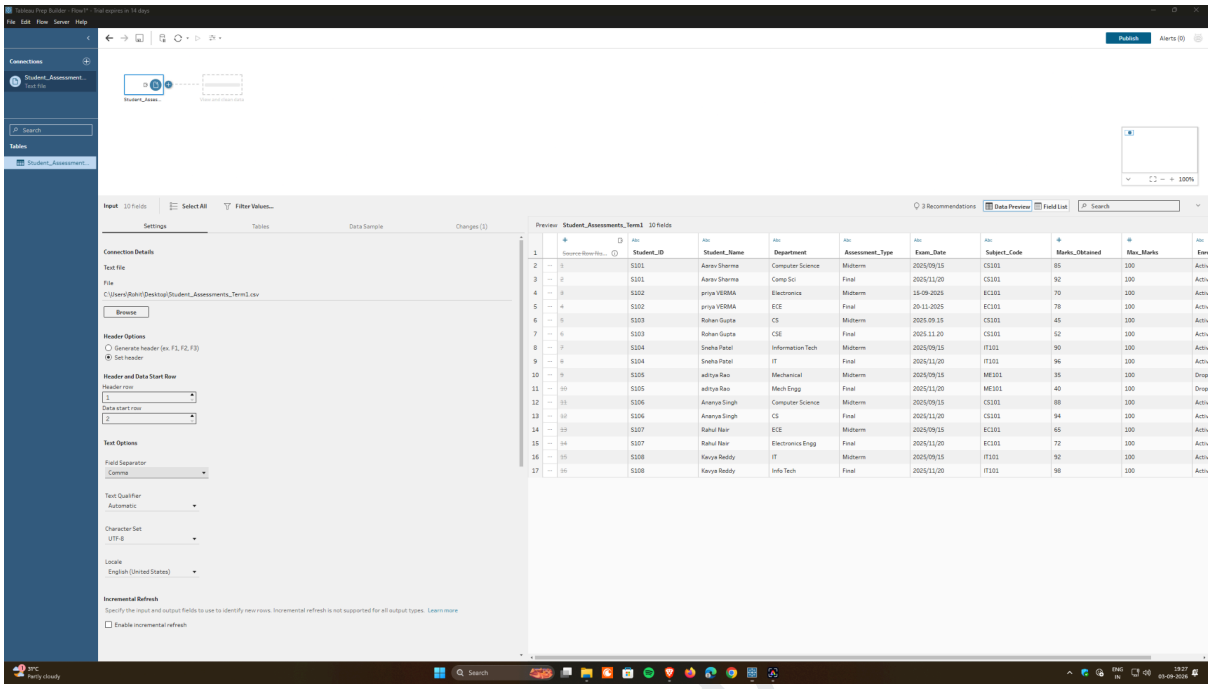


Select **Text file**, browse to **Student_Assessments_Term1.csv**, click **Open**, then verify the **Student_Assessments_Term1** Input Step appears and **Field Separator** is set to **Comma (,)**.

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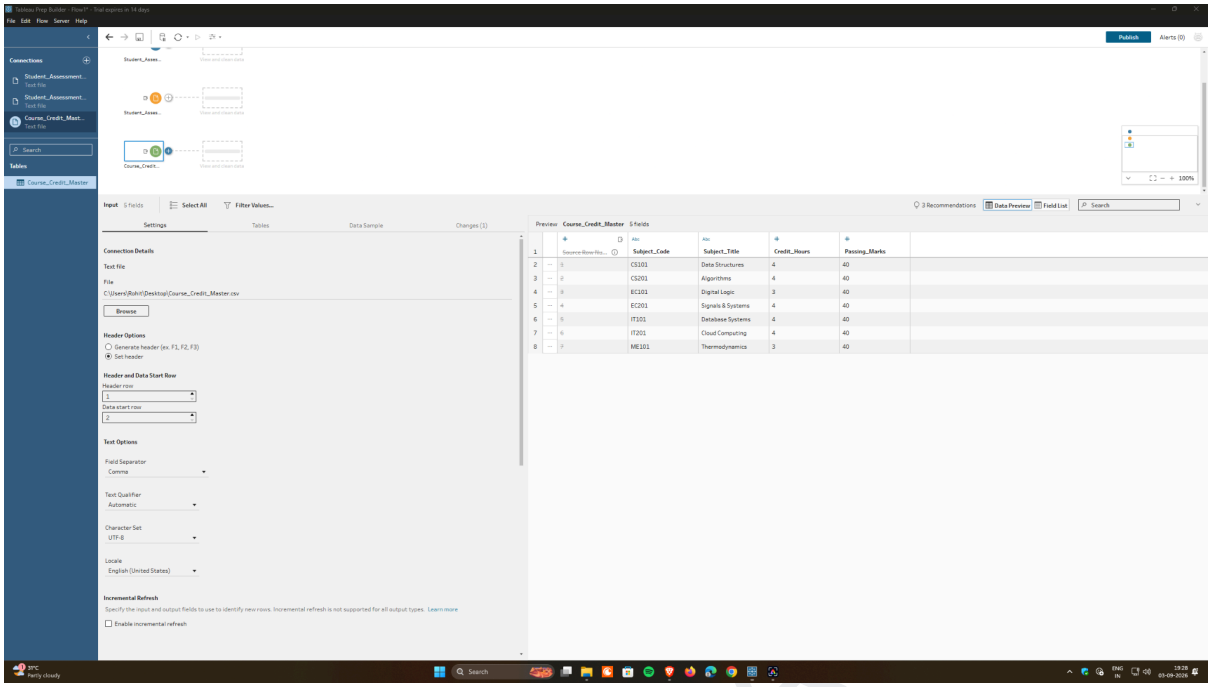
Step 3: Click **+ (Add Connection)** twice, select **Text file** each time, and open **Student_Assessments_Term2.csv** and **Course_Credit_Master.csv**, resulting in three separate **Input** icons on the Flow canvas.

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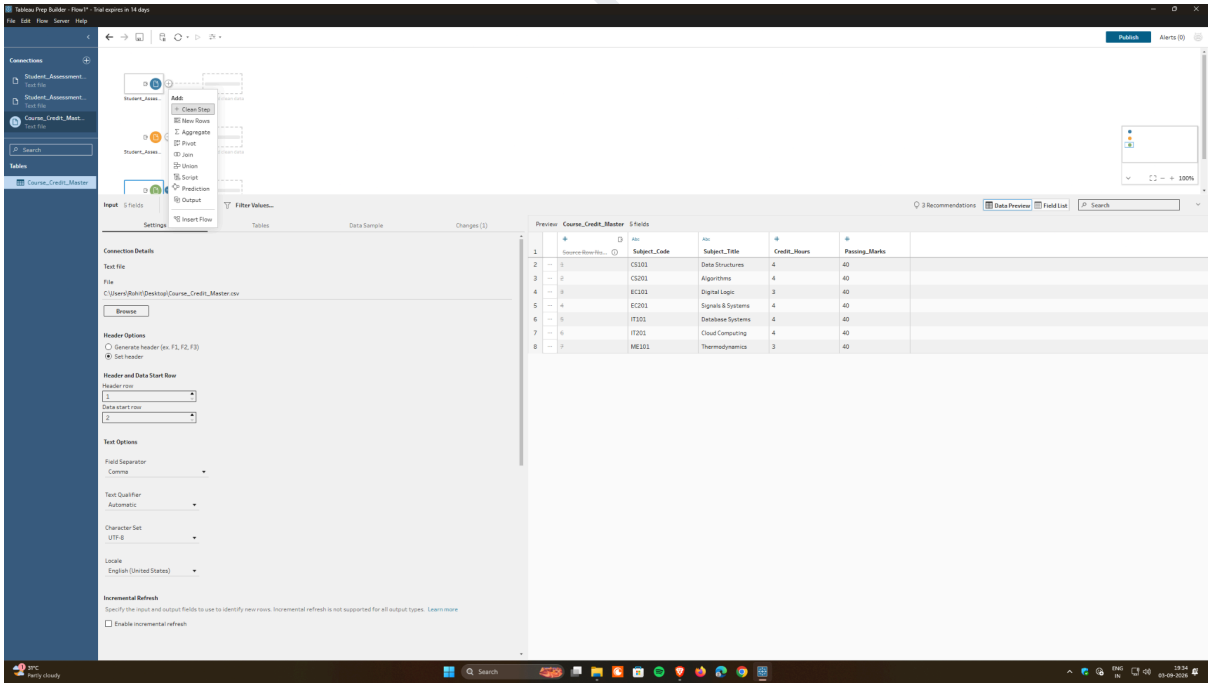
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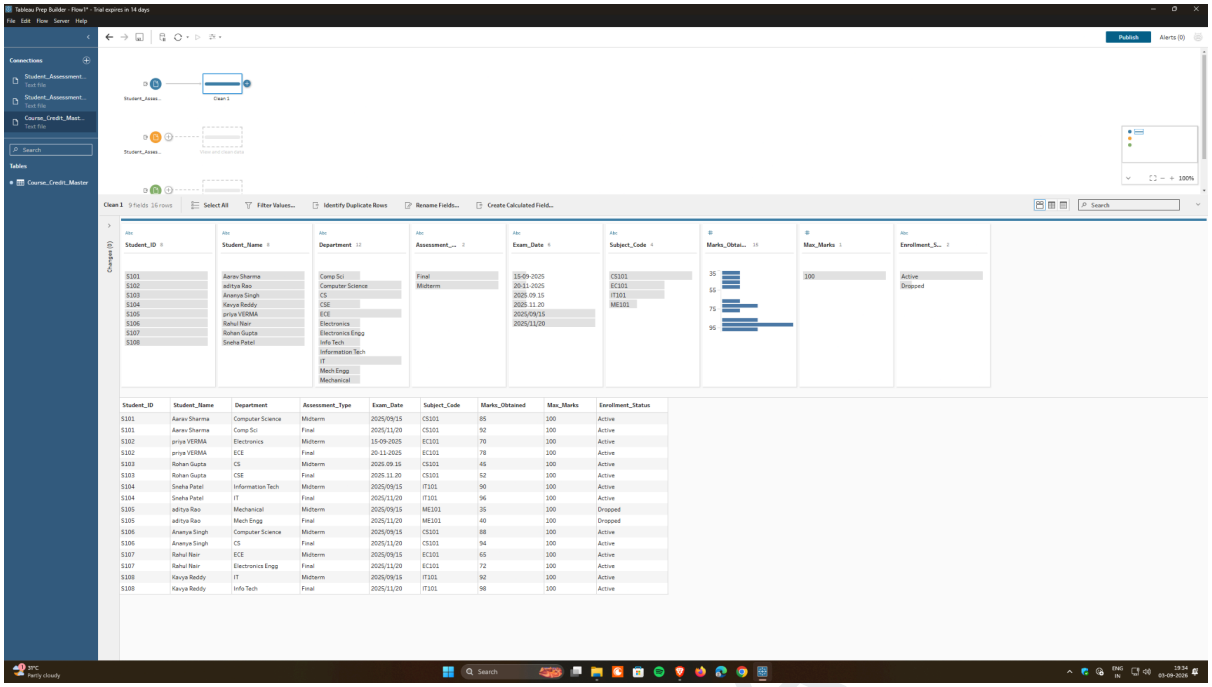
Step 4: Hover over **Student_Assessments_Term1**, click the + icon, select **Clean Step**, and verify that the **Profile Pane** opens showing cards for all **9 columns**.



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B: Clean and Profile Data

Objective: Use Tableau Prep Builder's interactive Profile Pane to identify data anomalies, whitespace issues, date formatting discrepancies, and inconsistent casing, then apply standard cleaning operations.

Unlike traditional spreadsheets, Tableau Prep provides real-time profiling cards that visualize distributions, frequency counts, and metadata data types for every field. Profiling allows you to quickly detect hidden formatting defects (such as extra spaces in names like ' Aarav Sharma ' or non-standard date strings) before standardizing categorical groups or performing joins.

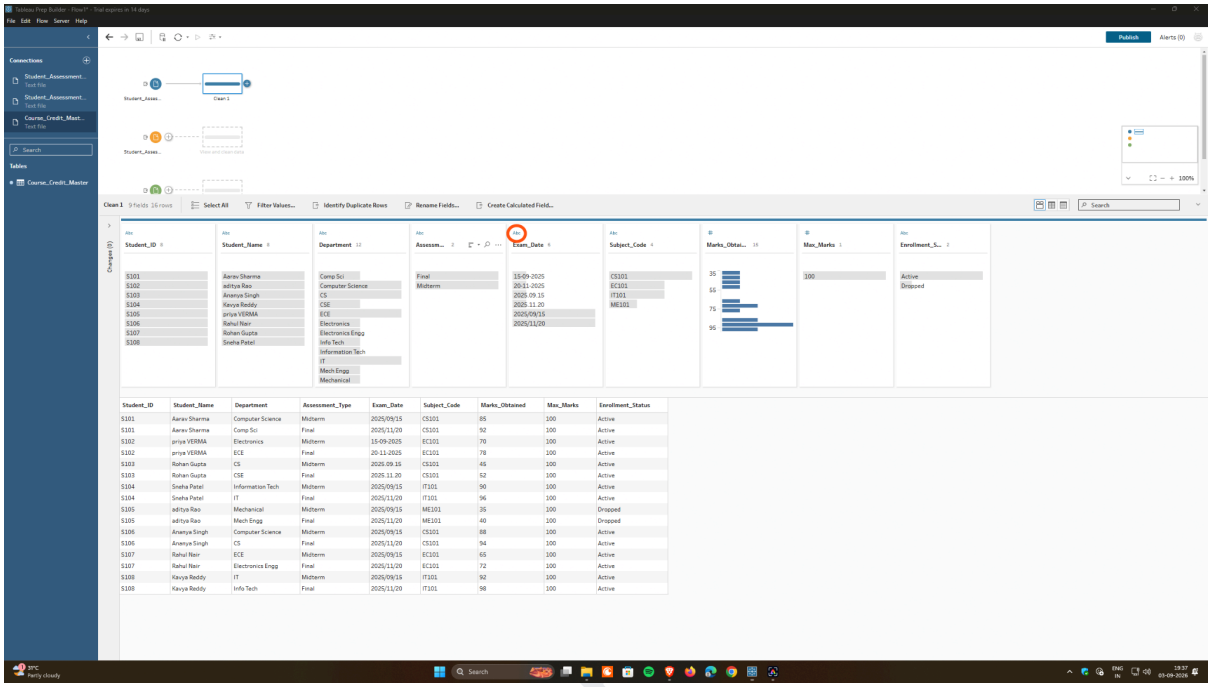
Step 1: Select the **Clean 1** step and inspect the **Profile Pane** to identify inconsistent **Student_Name** values (extra spaces/mixed casing) and **Exam_Date** stored as a string due to mixed separators.
t Exam_Date is recognized as a string (Abc) due to mixed separators
(/, -, .).

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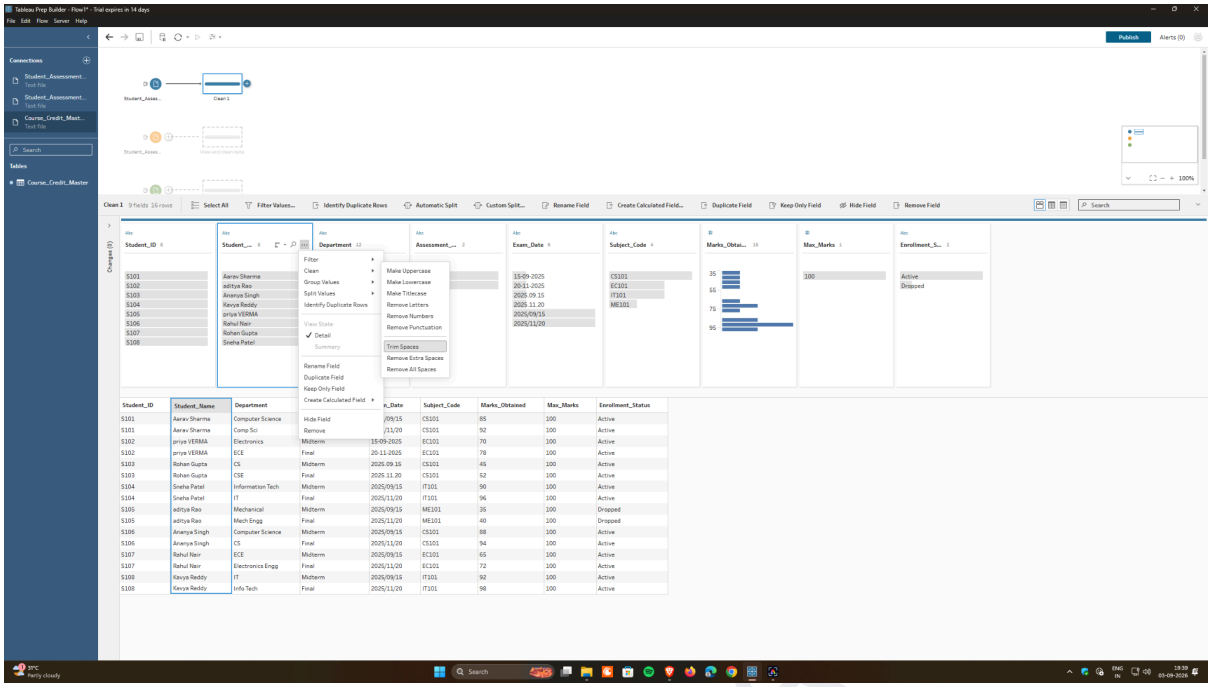
Step 2: On the Student_Name card, select More options (...) → Clean → Trim Spaces, then repeat with Make Uppercase to remove extra spaces and standardize the names.

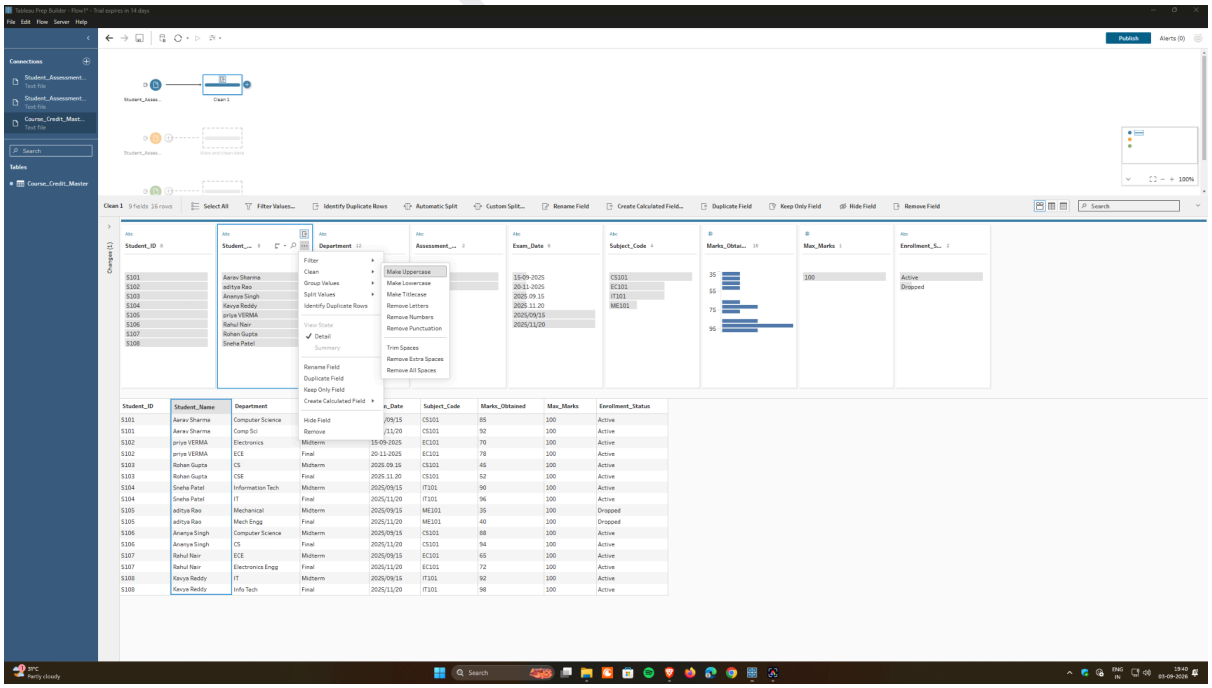
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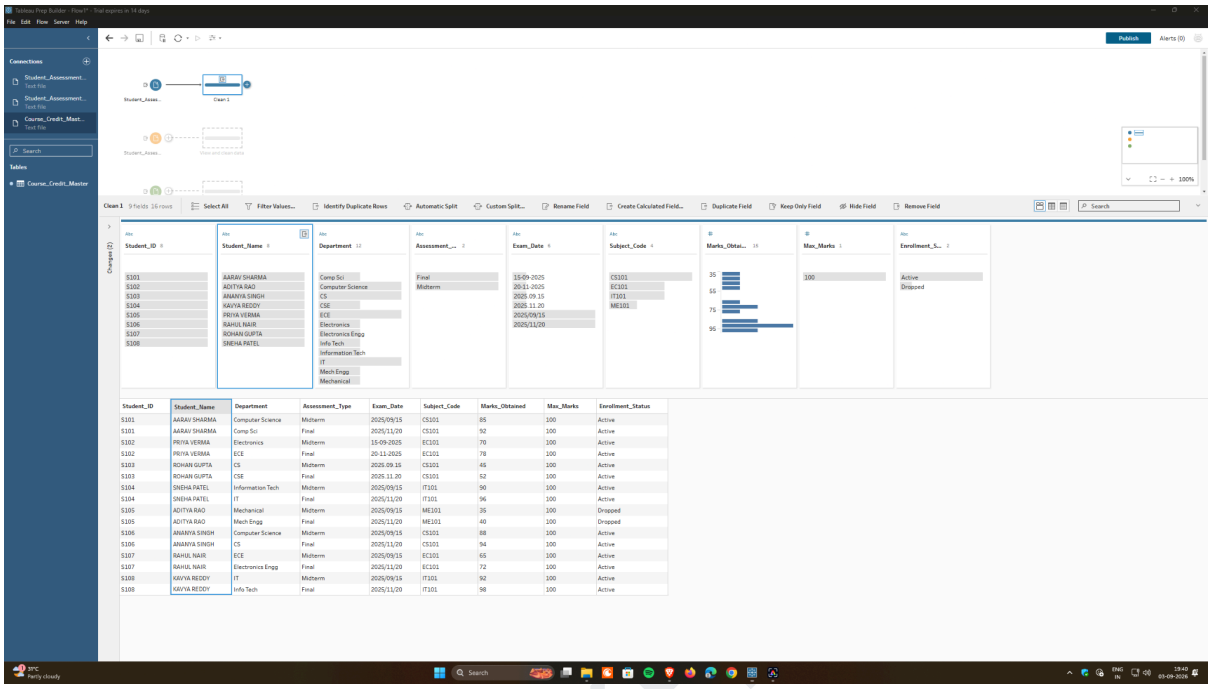




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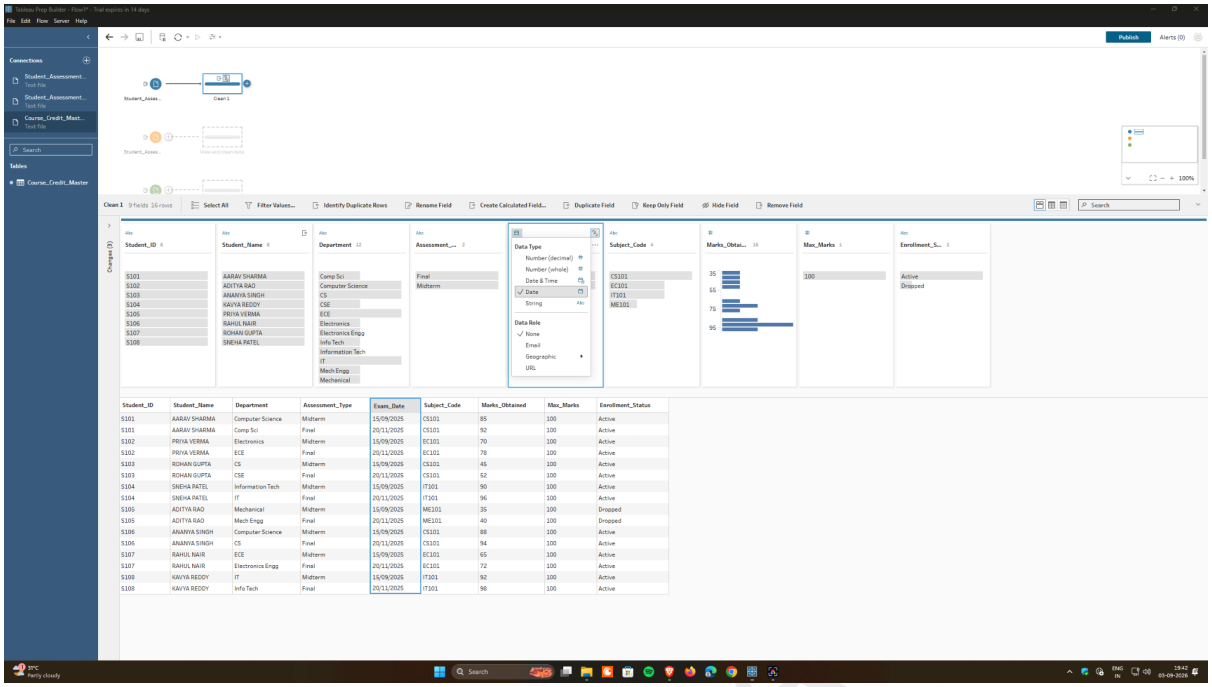
Step 3: On the **Exam_Date** card, click the **Abc data type icon**, select **Date**, and let Tableau Prep automatically standardize the mixed date formats into calendar dates.

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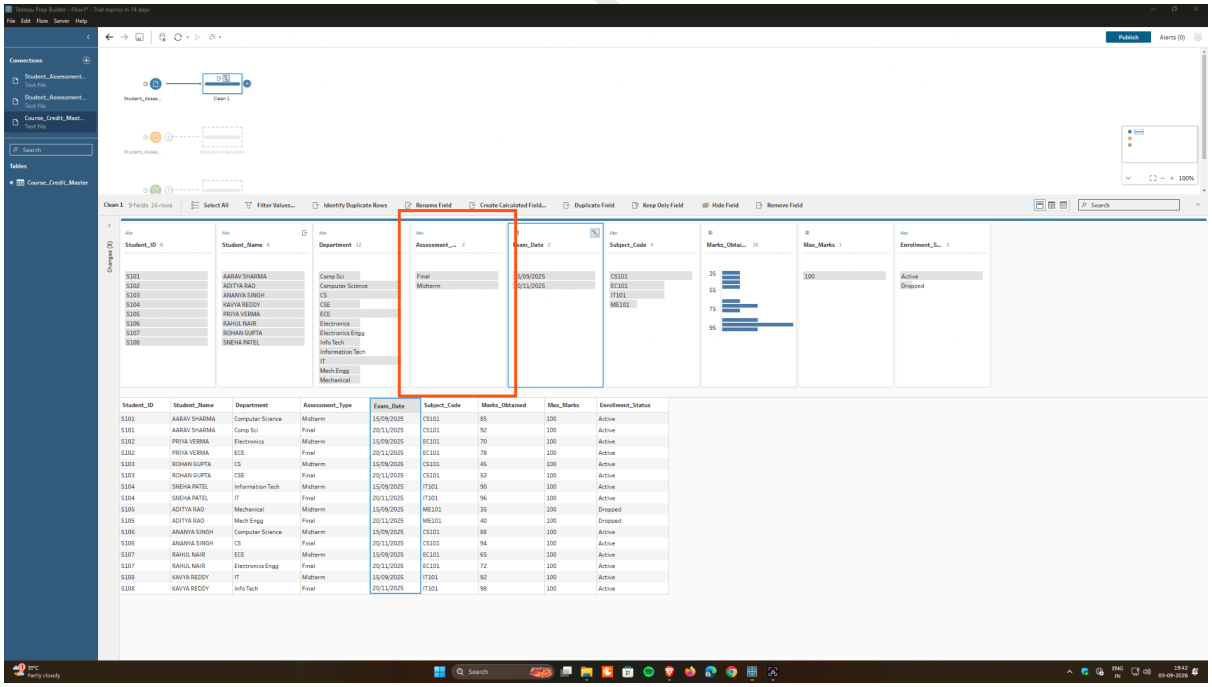
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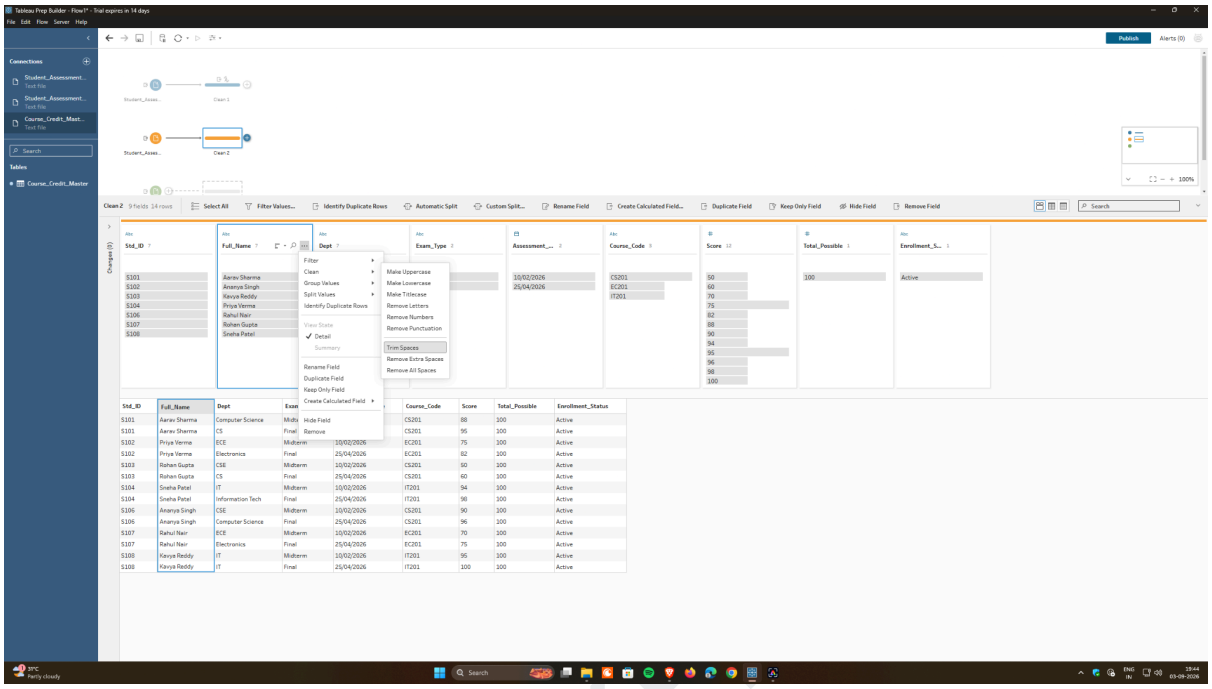


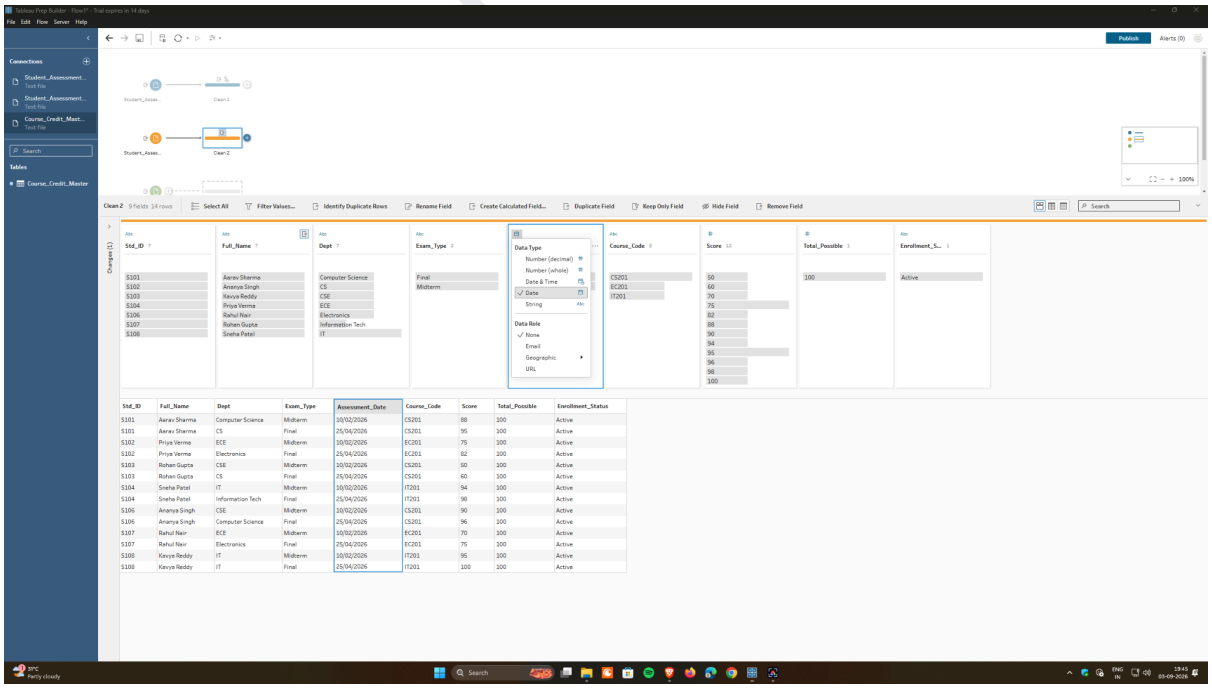
Step 4: Add a **Clean Step** to **Student_Assessments_Term2**, then in **Clean 2** apply **Trim Spaces** to **Full_Name** and change **Assessment_Date** from **Abc** to **Date**.

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Step 5: Open the **Changes** tab on the left to review the itemized, reversible transformations, such as **Trim Spaces** and **data type changes**.

Profile & Cleaning Audit Table:

Field Name	Detected Issue	Cleaning Operation Applied	Target State
Student_Name	Leading/trailing spaces (' Aarav Sharma ')	Clean Trim Spaces	Single trimaligned name
Student_Name	Inconsistent casing ('priya VERMA')	Clean Make Uppercase	Standardized casing
Exam_Date	Text datatype with mixed separators	Change Data Type Date	Standard Date format (YYYYMM-DD)
Assessment_Date	String data type	Change Data Type Date	Standard Date format (YYYYMM-DD)

C: Group and Standardize Values

Objective: Standardize spelling variations, abbreviations, and synonyms in categorical columns (such as Department and Dept) into clean, consistent values using Tableau Prep's built-in grouping algorithms.

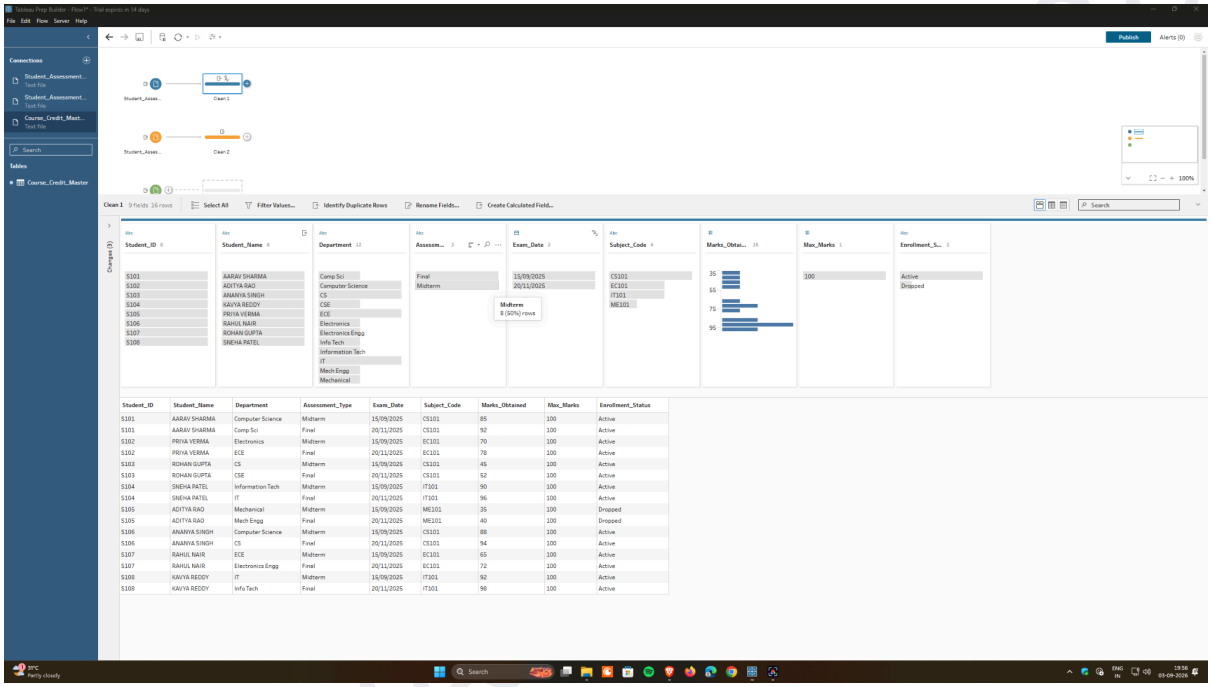
In academic datasets, departments are frequently entered under different aliases (e.g., "Comp Sci", "CS", "CSE", and "Computer Science"). If left unstandardized, downstream visual analysis will treat each alias as a completely different academic department, splintering the summary metrics.

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Step 1: IIn **Clean 1**, locate the **Department** card and identify the 12 distinct entries that represent 4 actual departments: **Computer Science, Electronics, Information Technology, and Mechanical Engineering**.



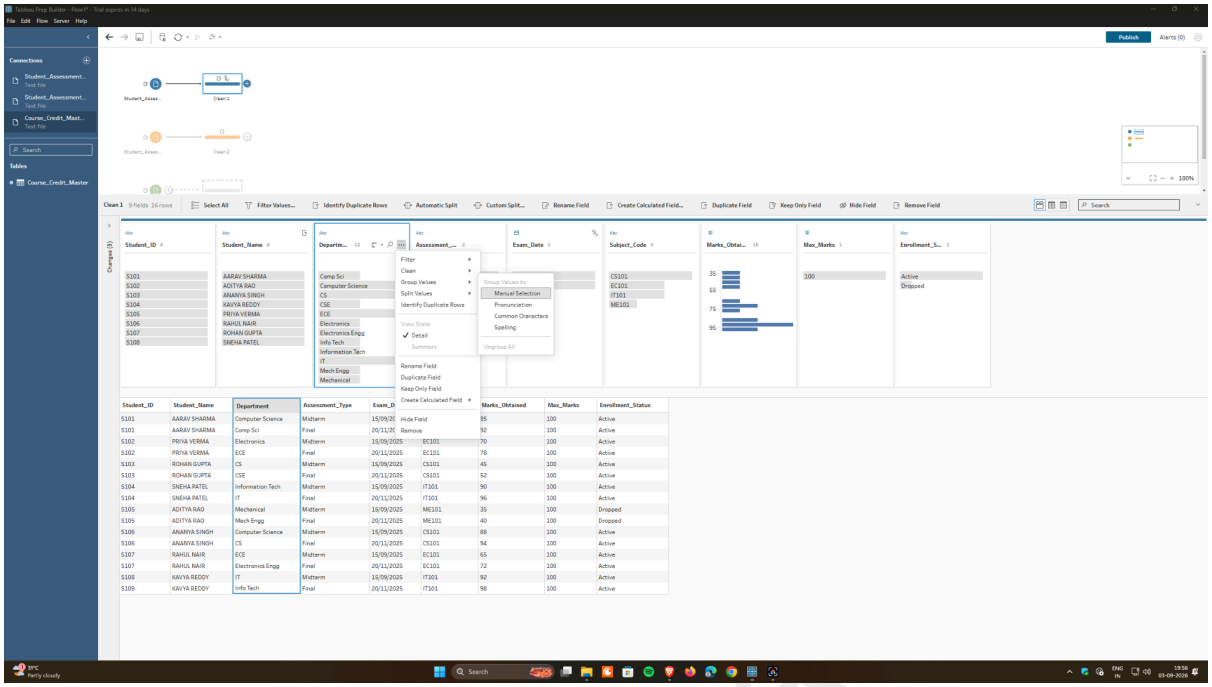
Step 2: On the **Department** card, select **More options (...) → Group Values → Manual Selection**, choose **Computer Science**, then group **Comp Sci, CS, and CSE** with it.

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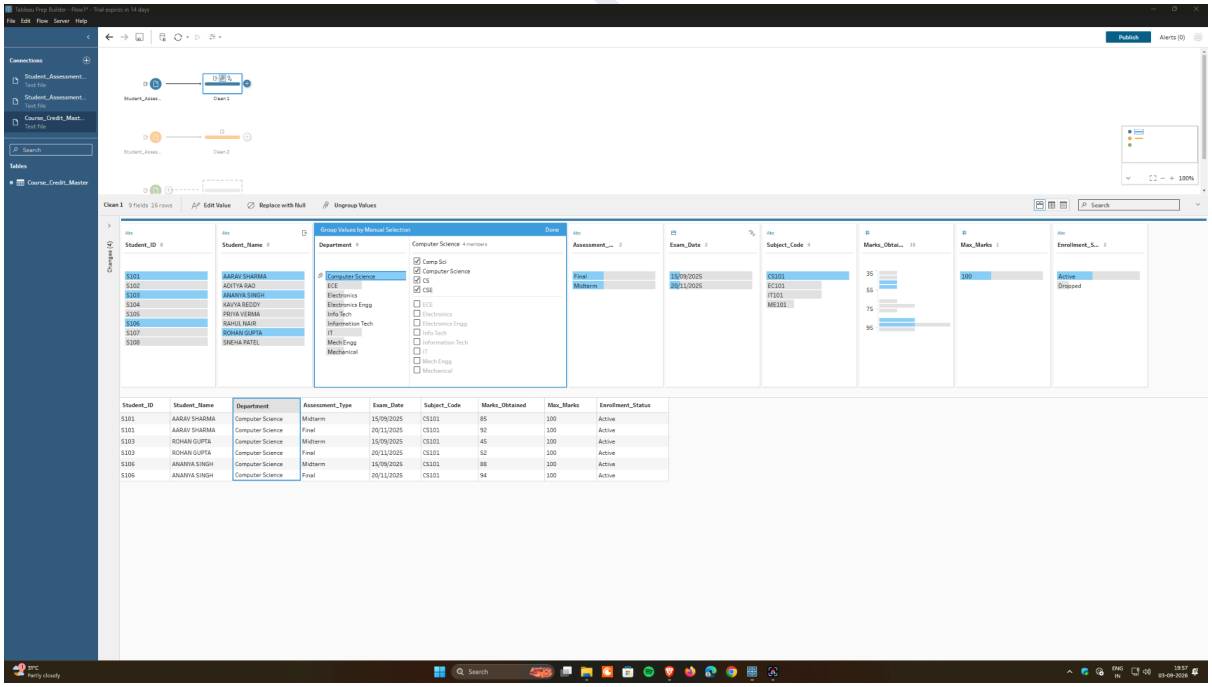
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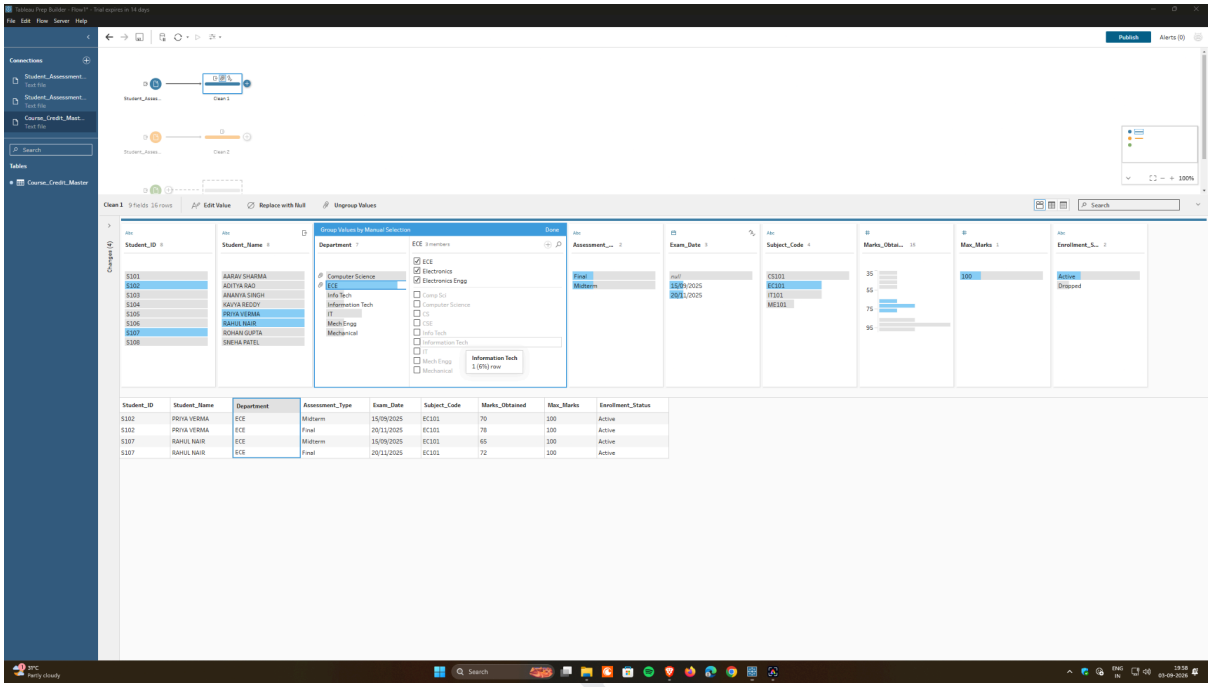


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Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.



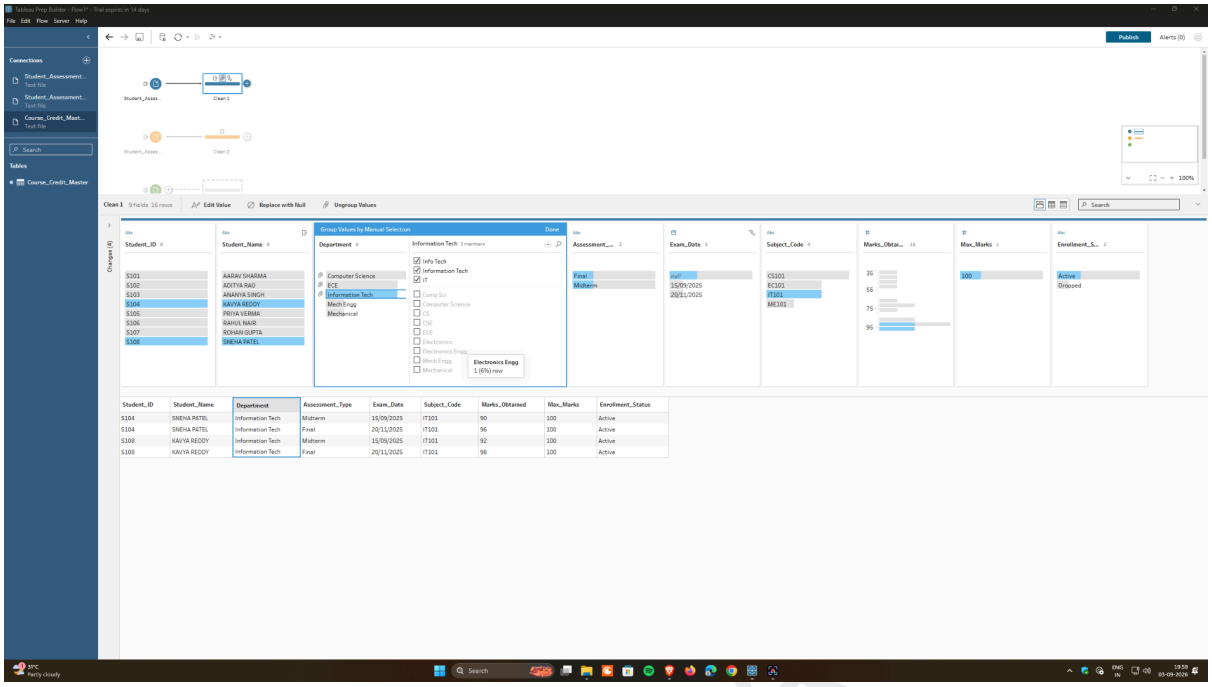
Click on Information Tech (or rename to Information Technology).Check the boxes for IT and Info Tech.

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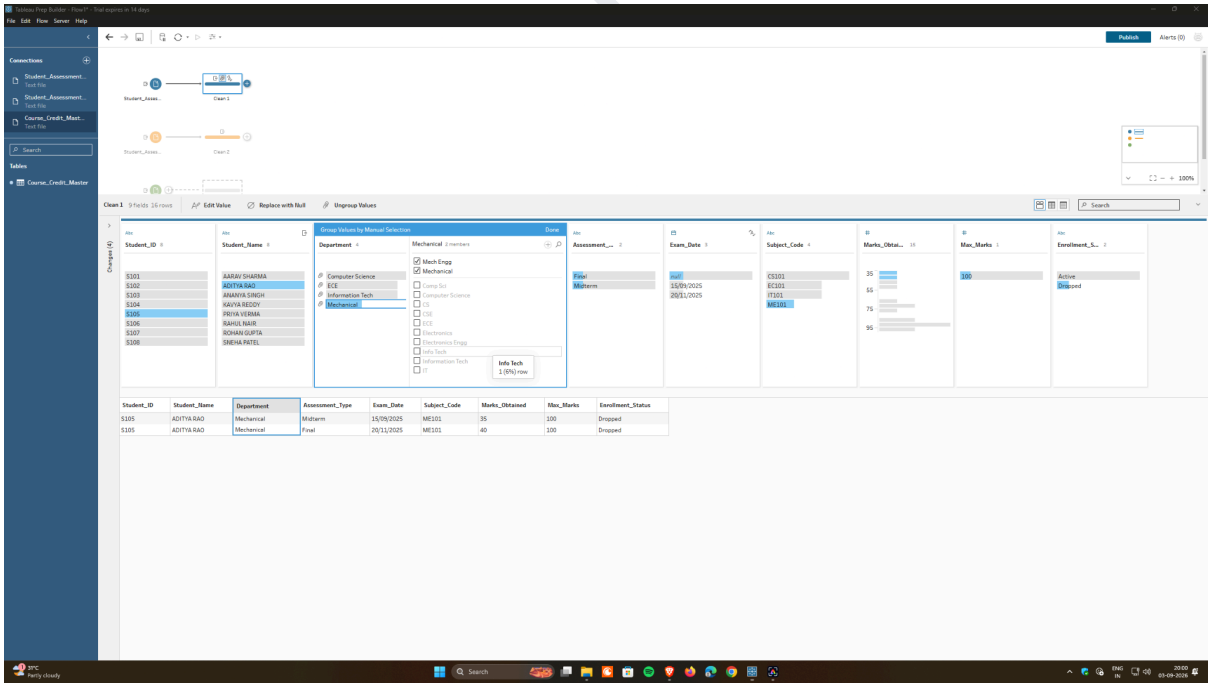
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Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.

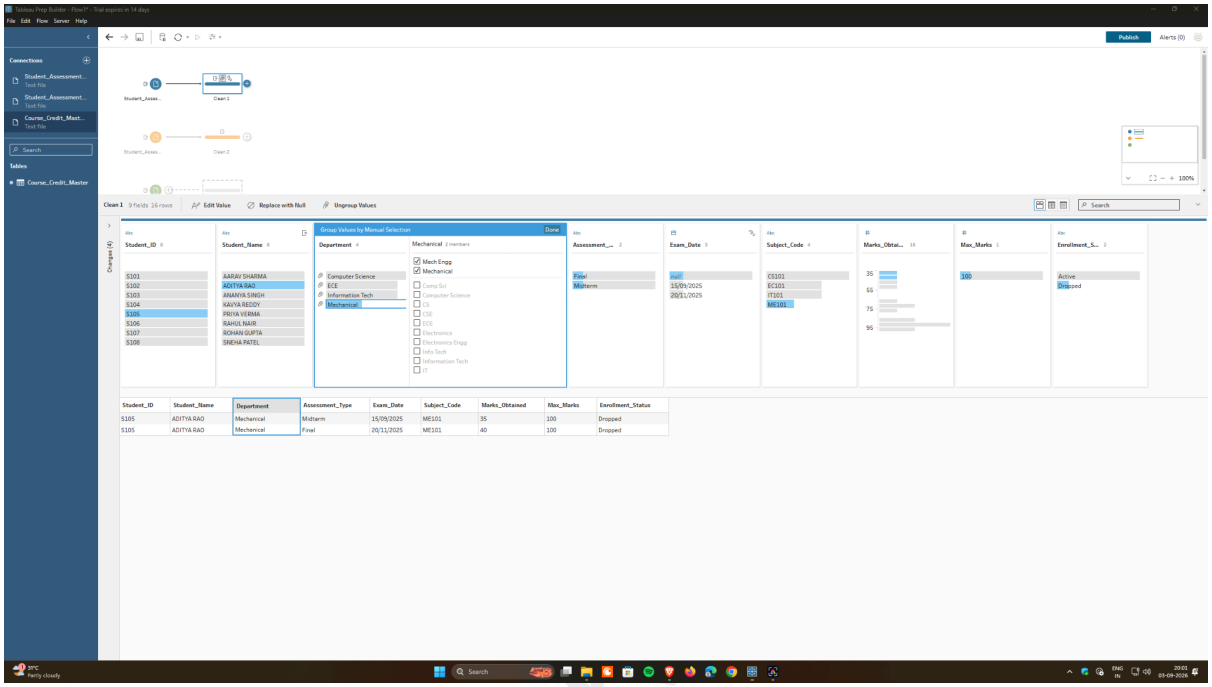


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Click **Done**, then verify the **paperclip icon** appears and the 12 messy department entries are consolidated into **4 clean departments**.



Step 3: In **Clean 2**, use **More options (...) → Group Values → Manual Selection** to group **CS, CSE, Computer Science → Computer Science; ECE, Electronics → Electronics; IT, Information Tech → Information Tech**, then click **Done**.

Standardization Mapping Table:

Original Input Variations	Target Standardized Value	Department Code
Computer Science, Comp Sci, CS, CSE	Computer Science	CS
Electronics, ECE, Electronics Engg	Electronics	EC
Information Tech, IT, Info Tech	Information Tech	IT

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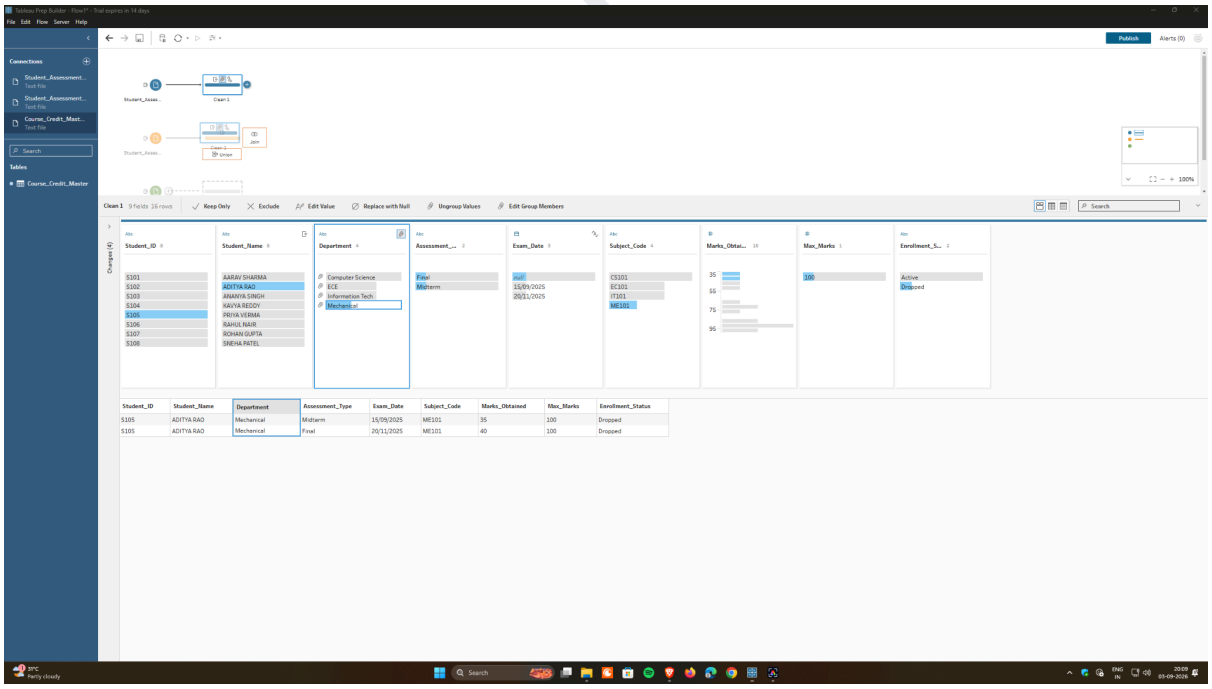
Mechanical, Mech Engg	Mechanical	ME

D: Merge Mismatched Fields

Objective: Perform a vertical Union of Clean 1 (Term 1) and Clean 2 (Term 2), resolve mismatched column headers (such as Std_ID vs Student_ID, Score vs Marks_Obtained), and unify them into a single schema.

When combining datasets originating from different semesters, departments, or source systems, identical information often resides under dissimilar column names. In Tableau Prep, unmerged fields create sparse datasets with duplicate null columns. Merging mismatched fields consolidates those columns without requiring upstream source file modifications.

Step 1: Drag **Clean 2** onto **Clean 1**, select **Union** from the drop zone, and verify that **Union 1** is created connecting both branches.

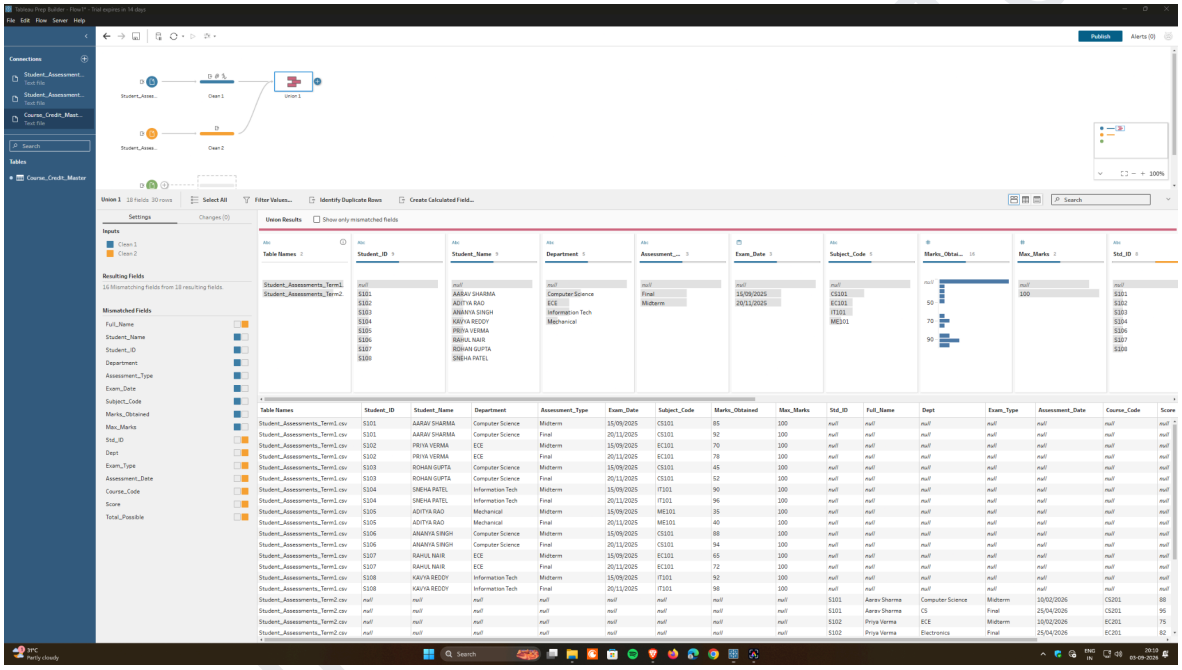


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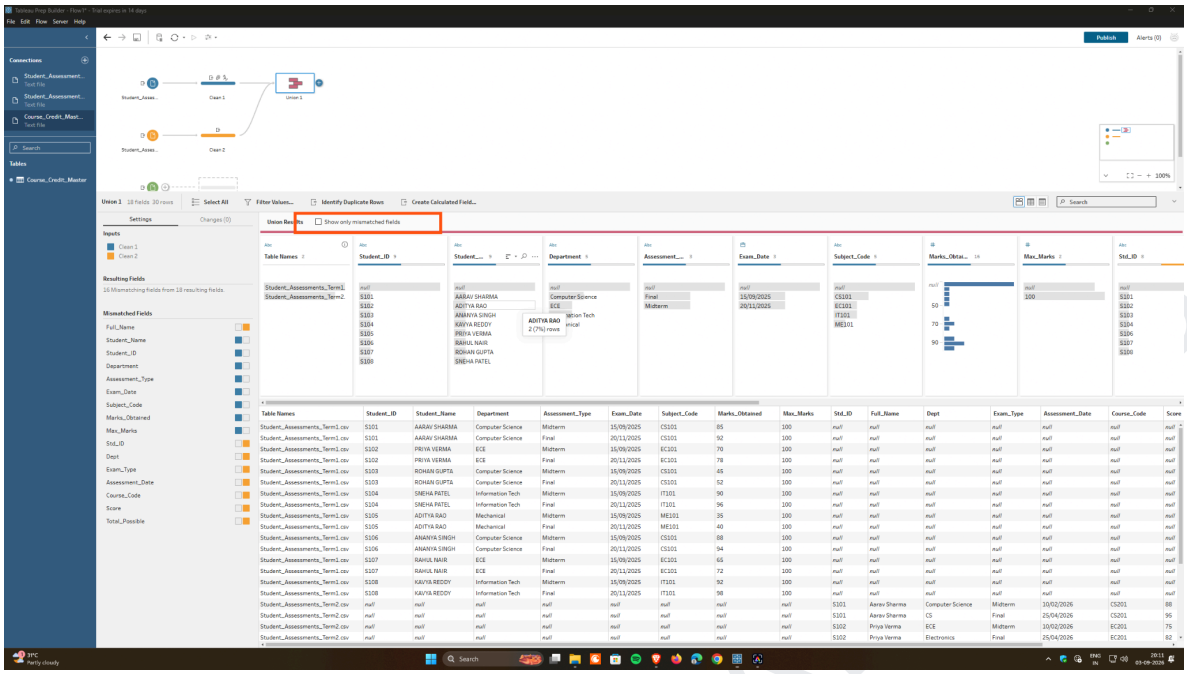


Step 2: Select **Union 1** and review the **Mismatched Fields** section to identify differently named but equivalent fields, such as **Student_ID/Std_ID, Student_Name/Full_Name, Department/Dept, Assessment_Type/Exam_Type, Exam_Date/Assessment_Date, Subject_Code/Course_Code, Marks_Obtained/Score, and Max_Marks/Total_Possible**.

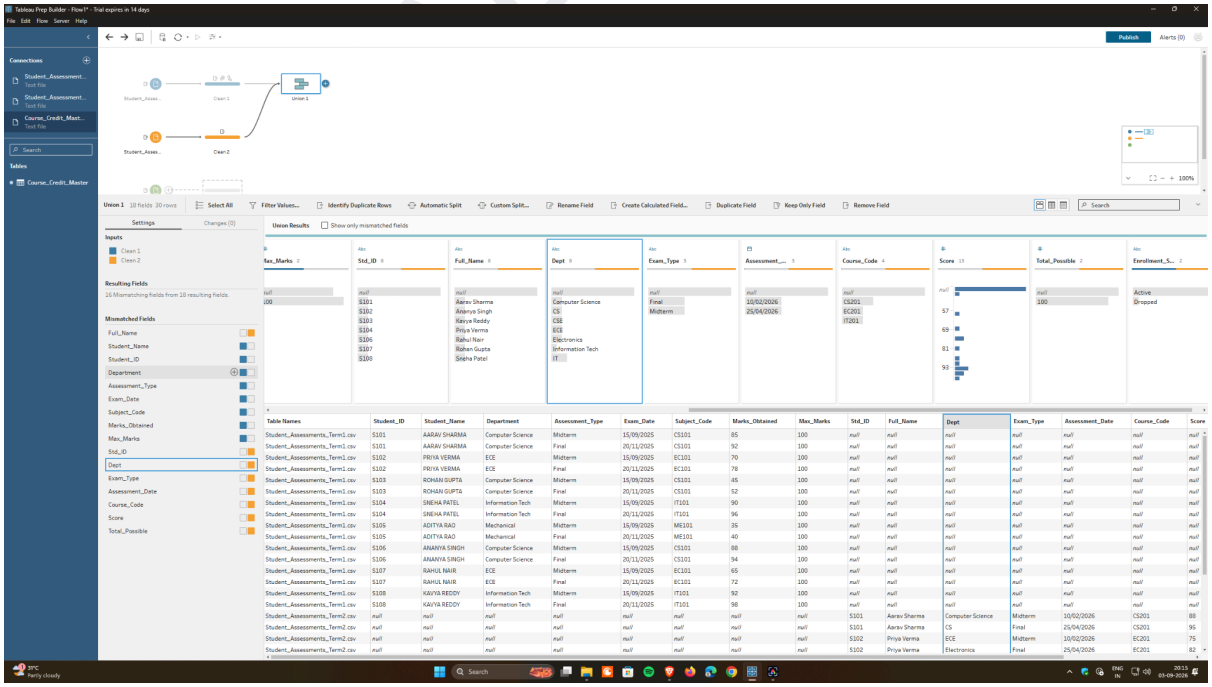
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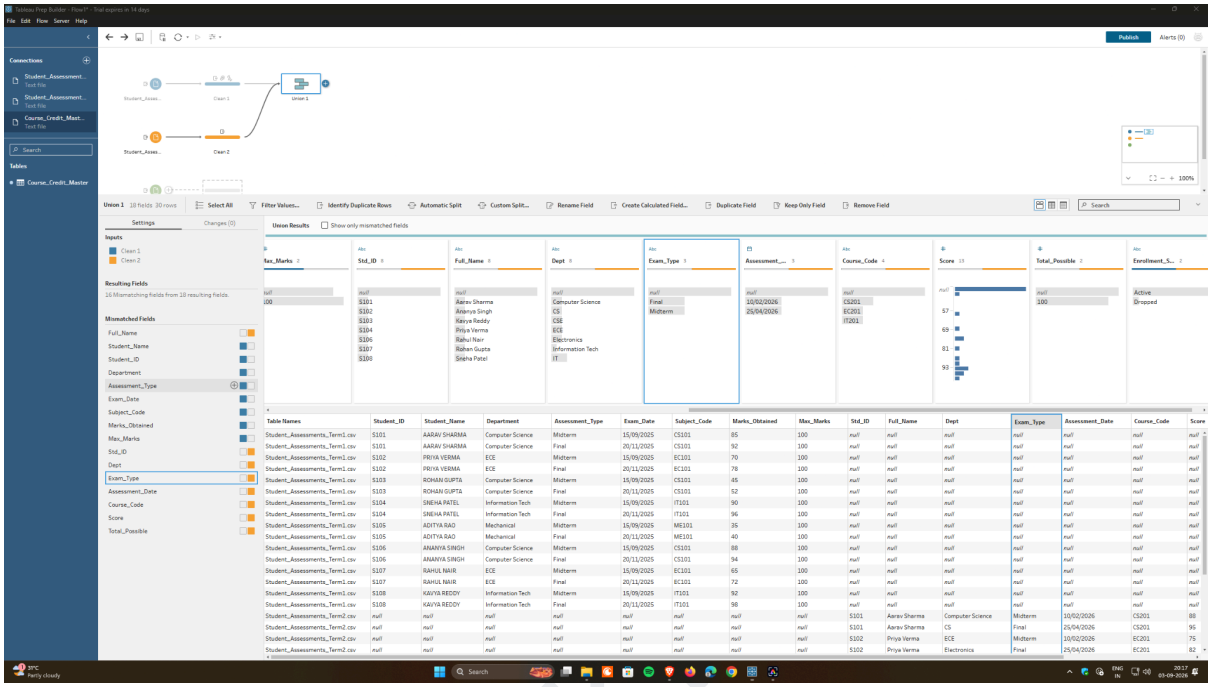
Step 3: Drag each mismatched field onto its corresponding field to merge **Std_ID→Student_ID, Full_Name→Student_Name, Dept→Department, Assessment_Date→Exam_Date, Course_Code→Subject_Code, Score→Marks_Obtained, Total_Possible→Max_Marks, and Exam_Type→Assessment_Type**, until the mismatch count becomes **0**.

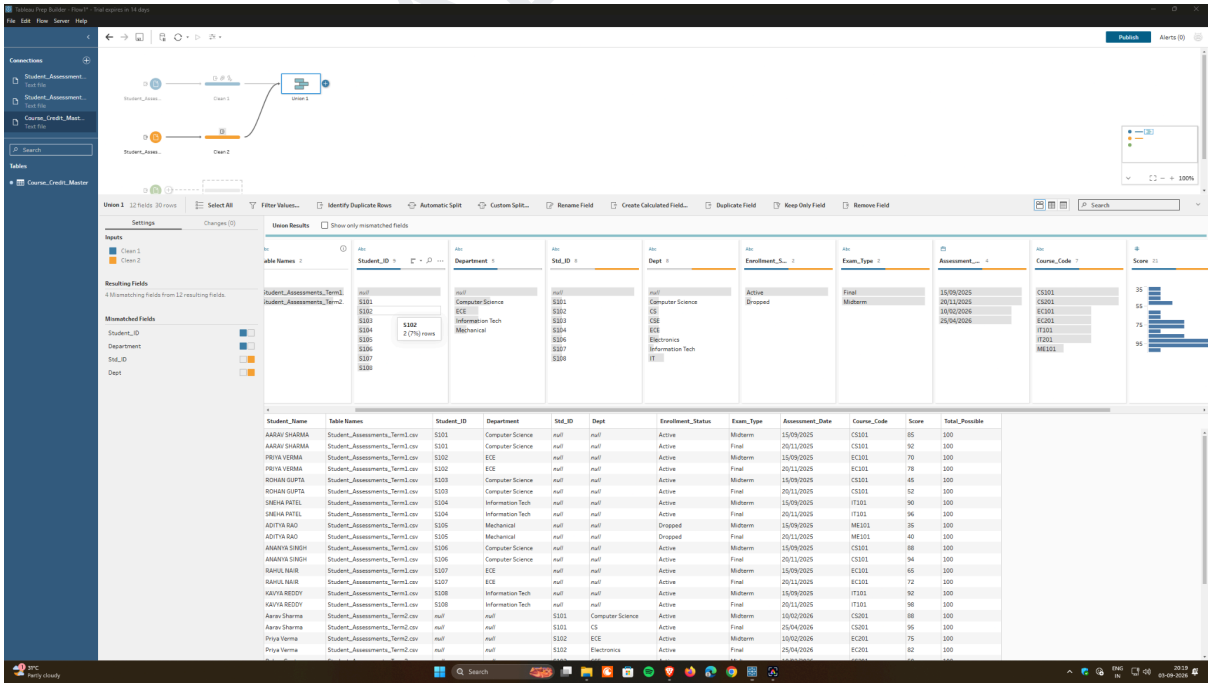


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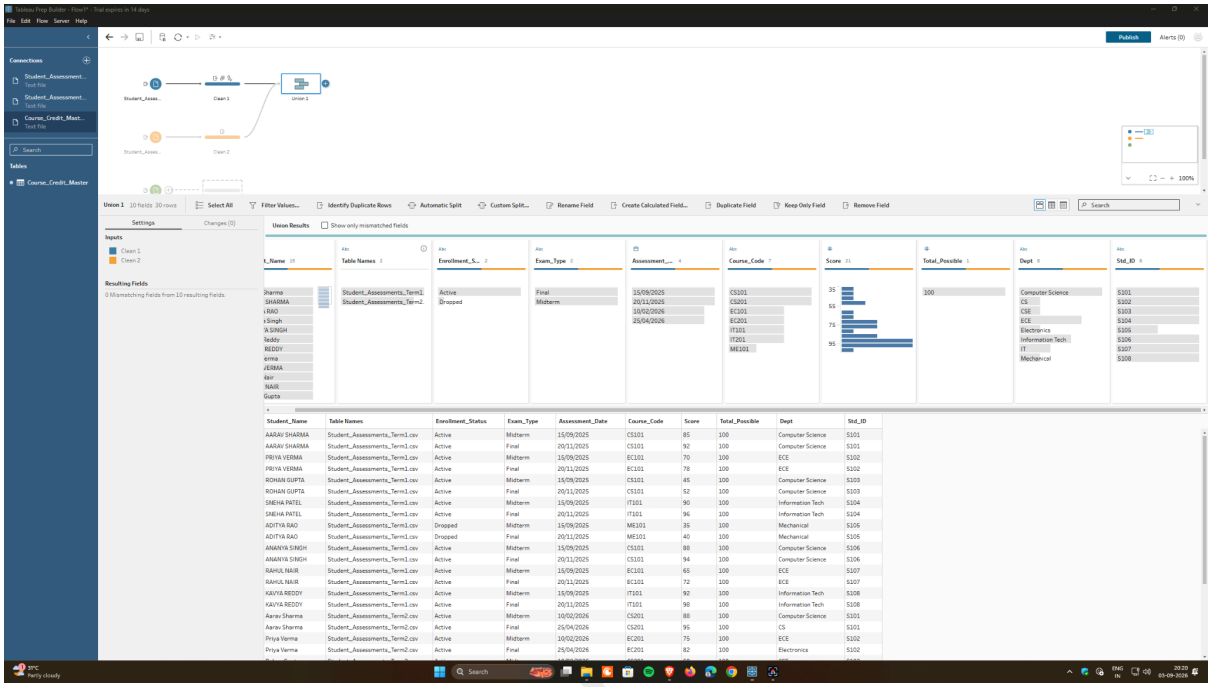




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Step 4: Click + next to **Union 1**, select **Clean Step** (creating **Clean 3**; it may appear as **Clean 4** if another Clean Step already exists), and verify **30 total rows** across **10 unified columns**, including **Table Names**

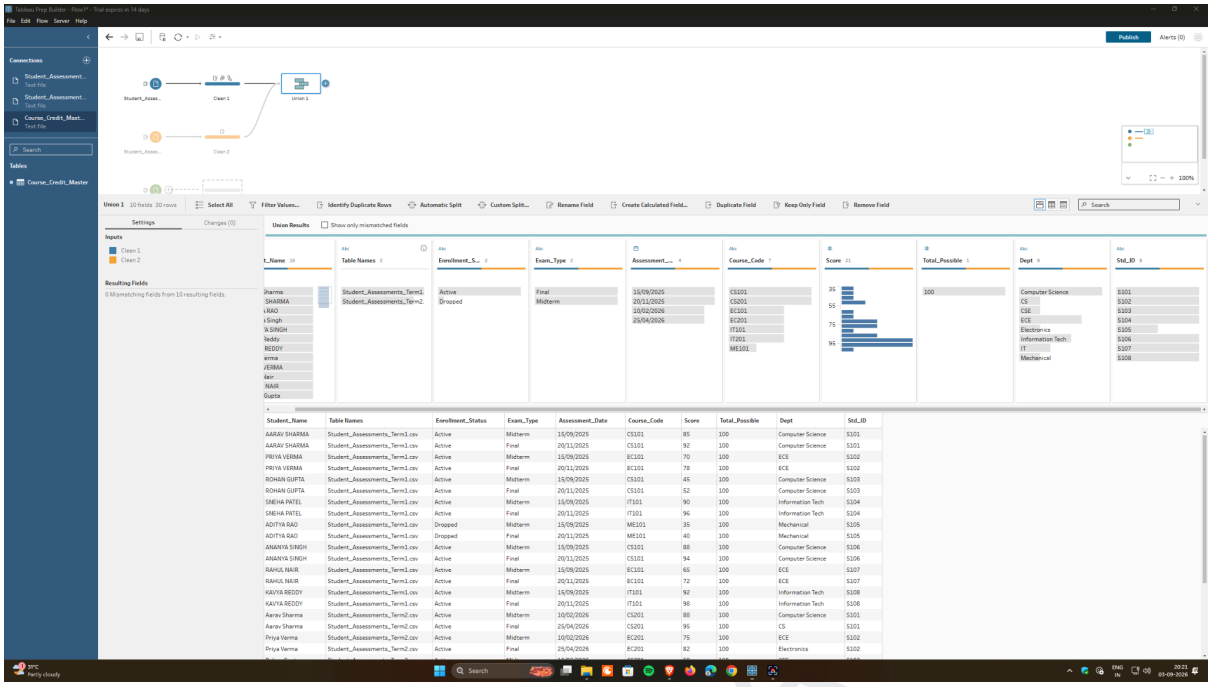
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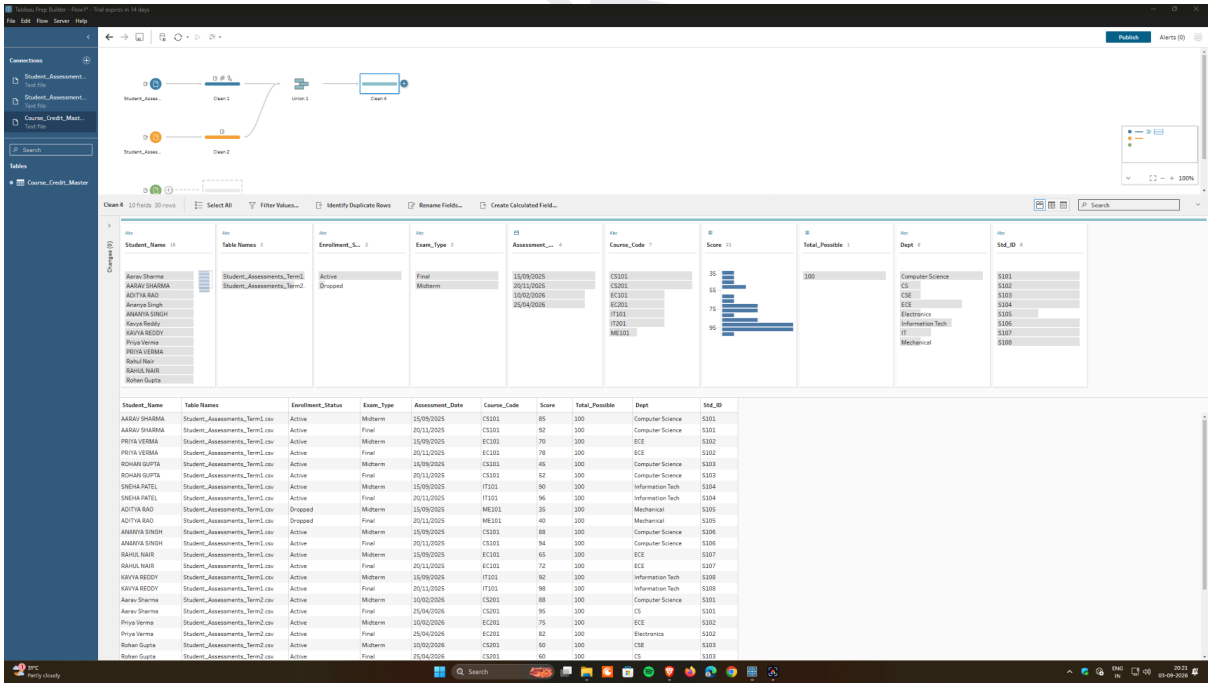
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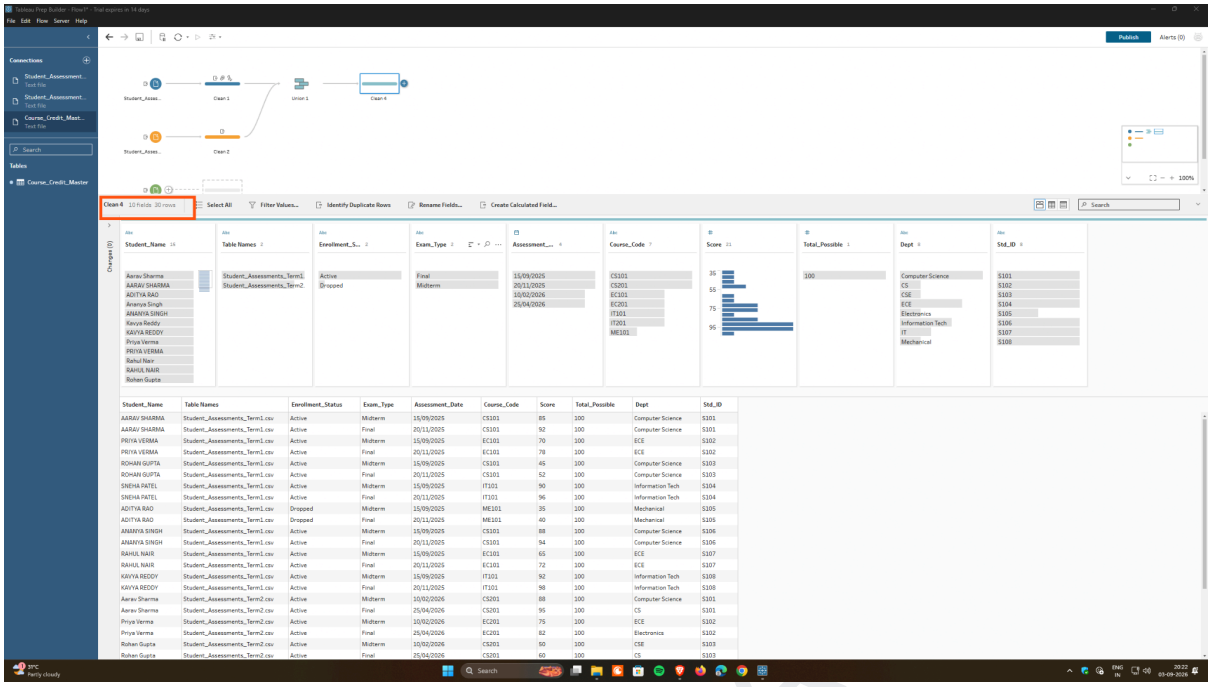




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E: Perform Aggregations in Prep

Objective: Create an Aggregate Step in Tableau Prep to summarize granular, exam-level test scores into student-level cumulative metrics (Total Marks, Average Score, and Total Assessment Count) per term

Raw assessment tables log every individual examination (Midterm, Final) on a separate row. By performing an aggregation step directly in Tableau Prep, we roll up individual row events into pre-computed summary statistics, reducing data redundancy and generating student scorecard baselines prior to dashboarding.

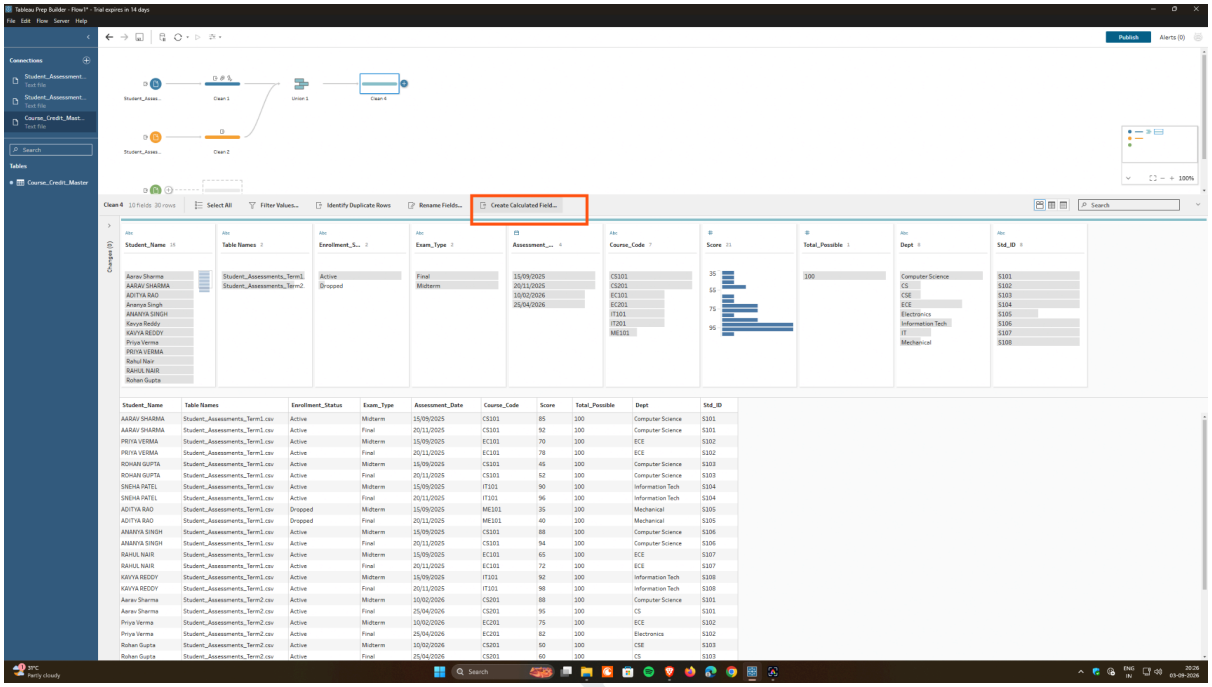
Step 1: In Clean 3, use More options (...) → Create Calculated Field, name it Term, enter the given `CONTAINS([Table Names], "Term1/Term2")` formula to classify records as Term 1, Term 2, or Other, then click Save.

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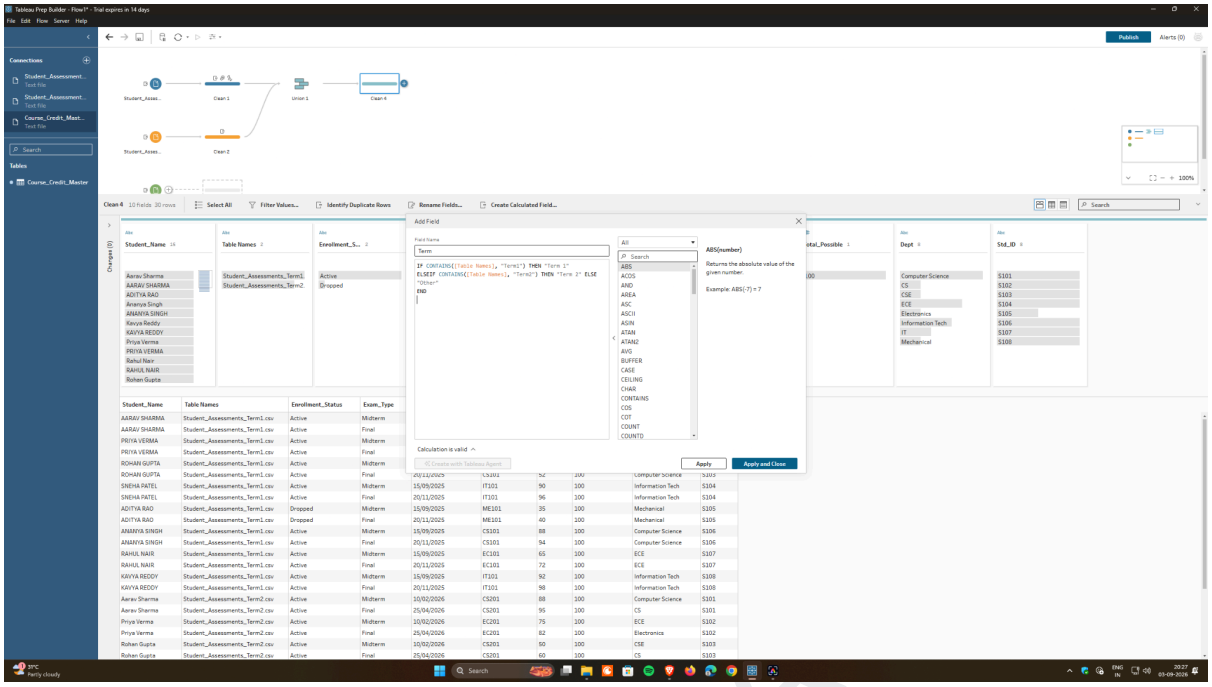


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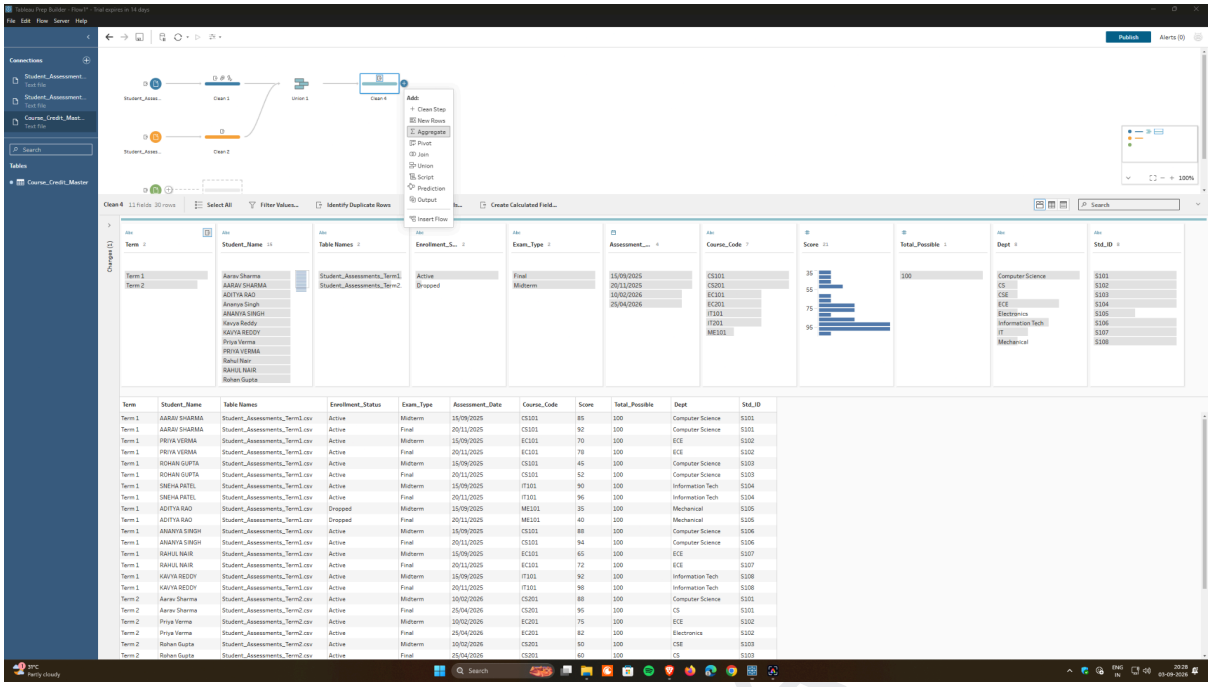
Step 2: Hover over Clean 3, click +, select Aggregate (∑), and verify that Aggregate 1 appears with Grouped Fields and Aggregated Fields drop zones.

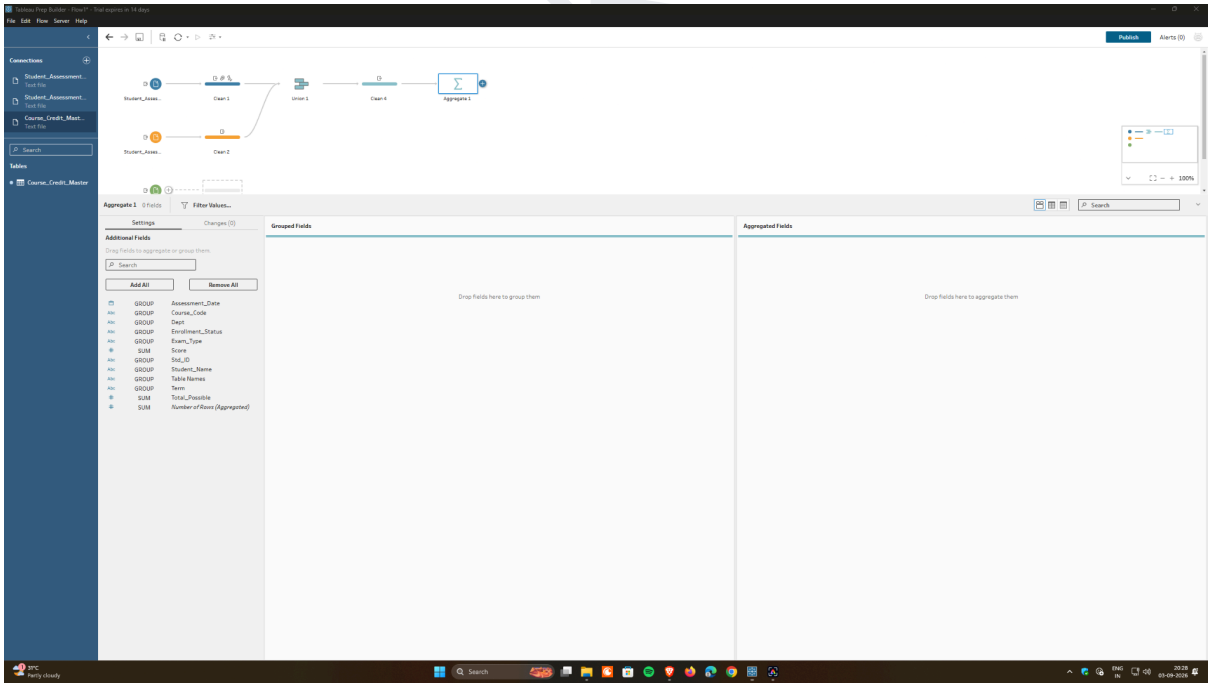
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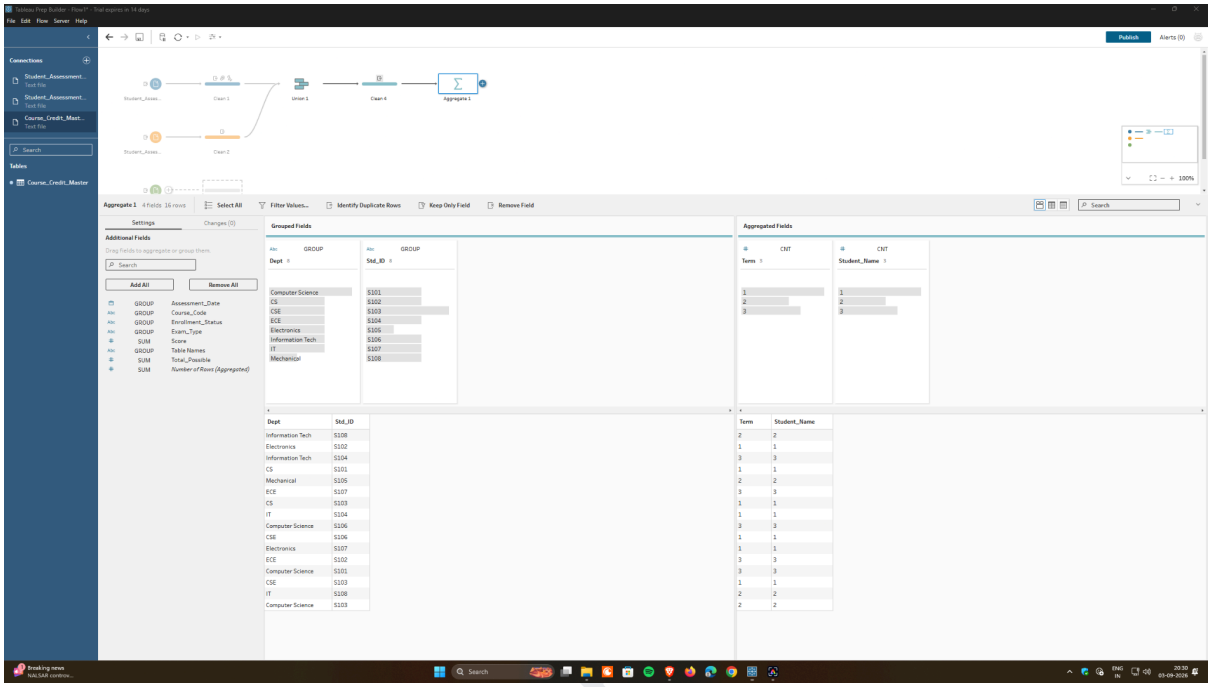


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Step 3: In the Aggregate step, drag Student_ID, Student_Name, Department, and Term into the Grouped Fields box.



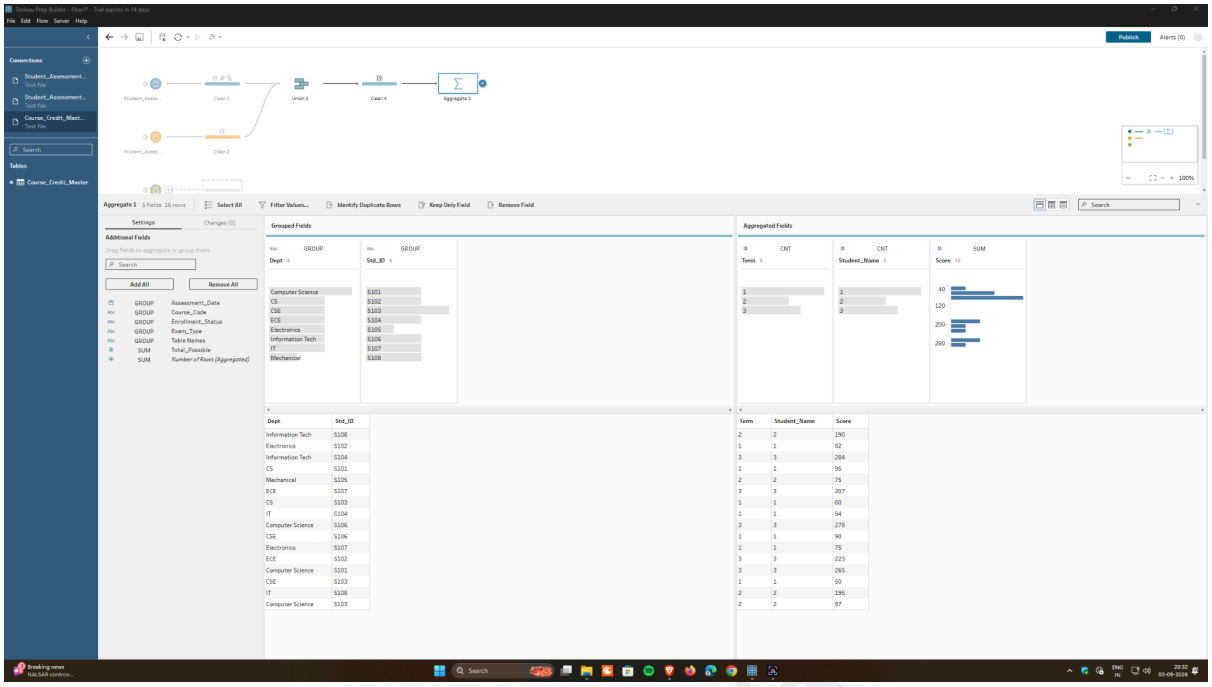
Step 4: In Aggregated Fields, add Marks_Obtained → Sum for Total Marks, use Marks_Obtained → Average for Average Score, and add Max_Marks → Sum for Total Possible Marks.

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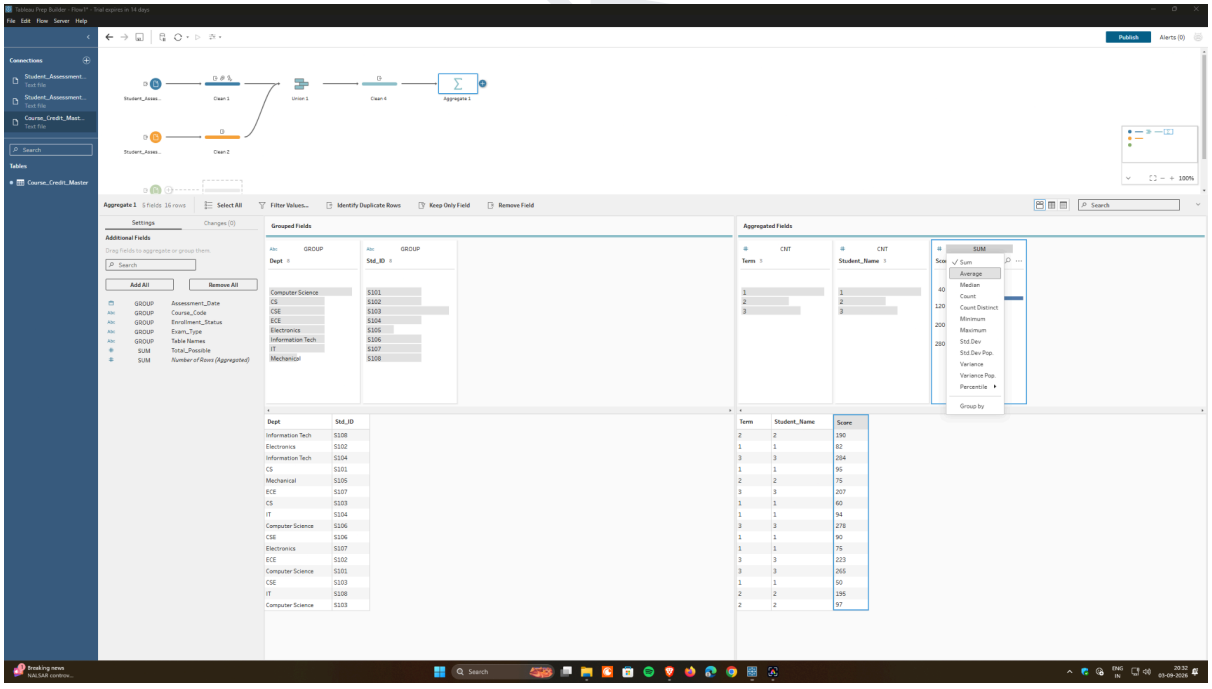
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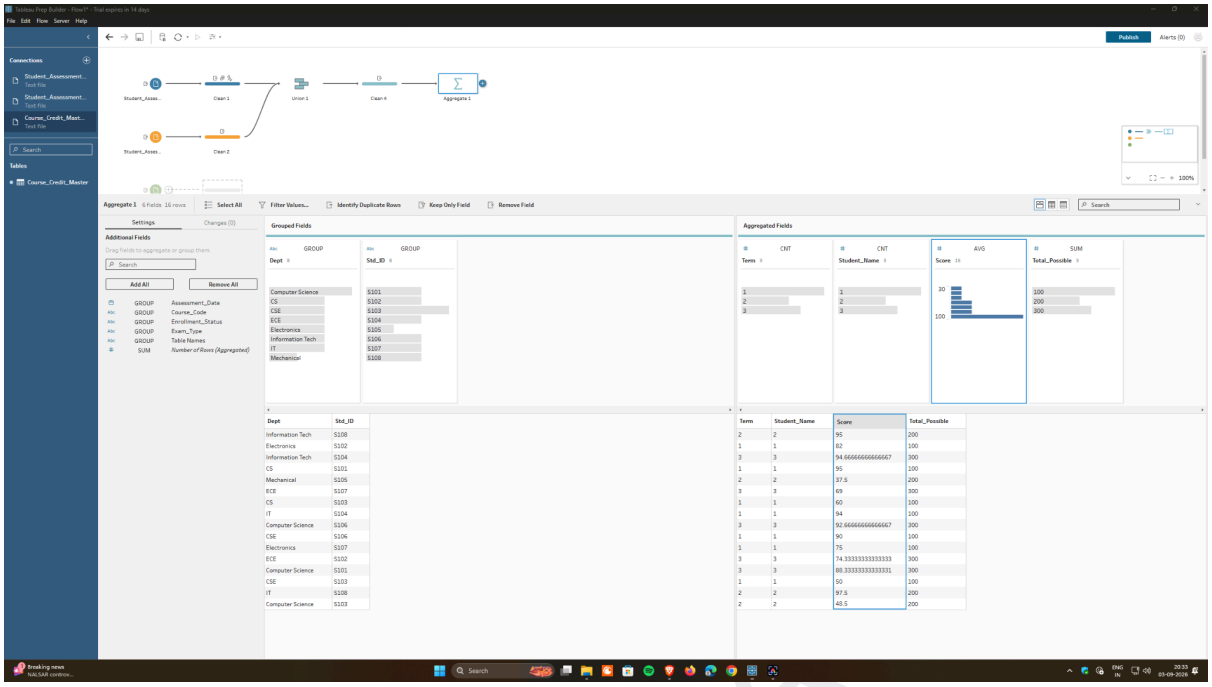
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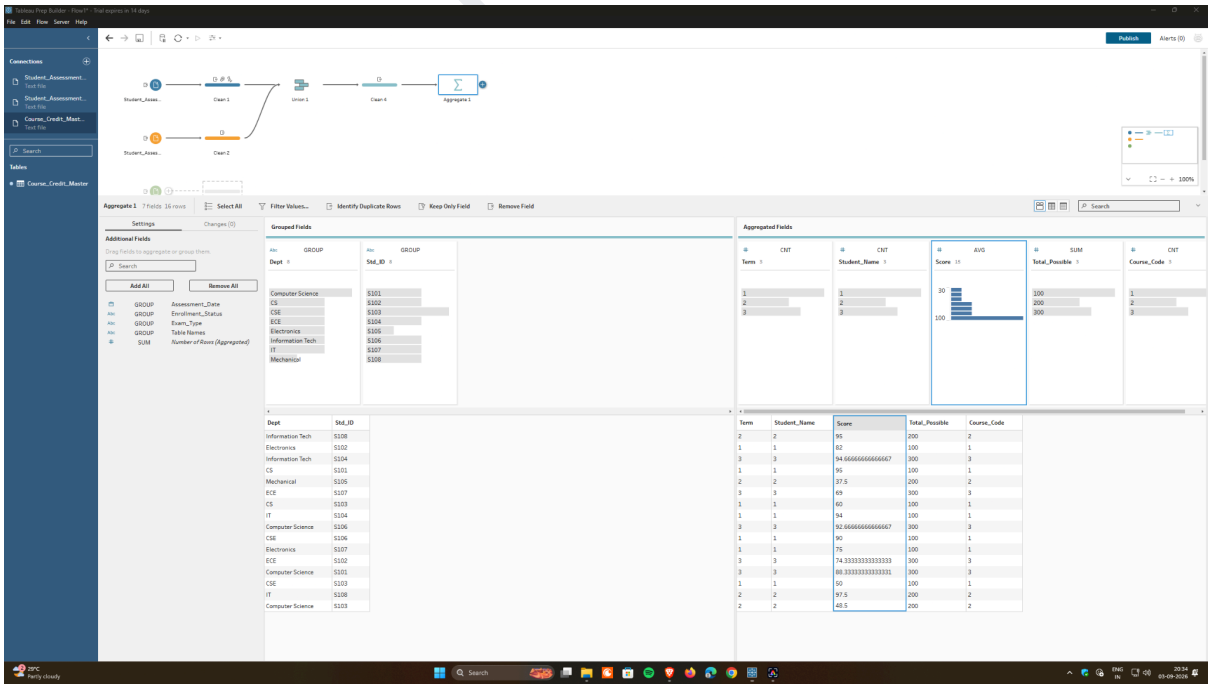


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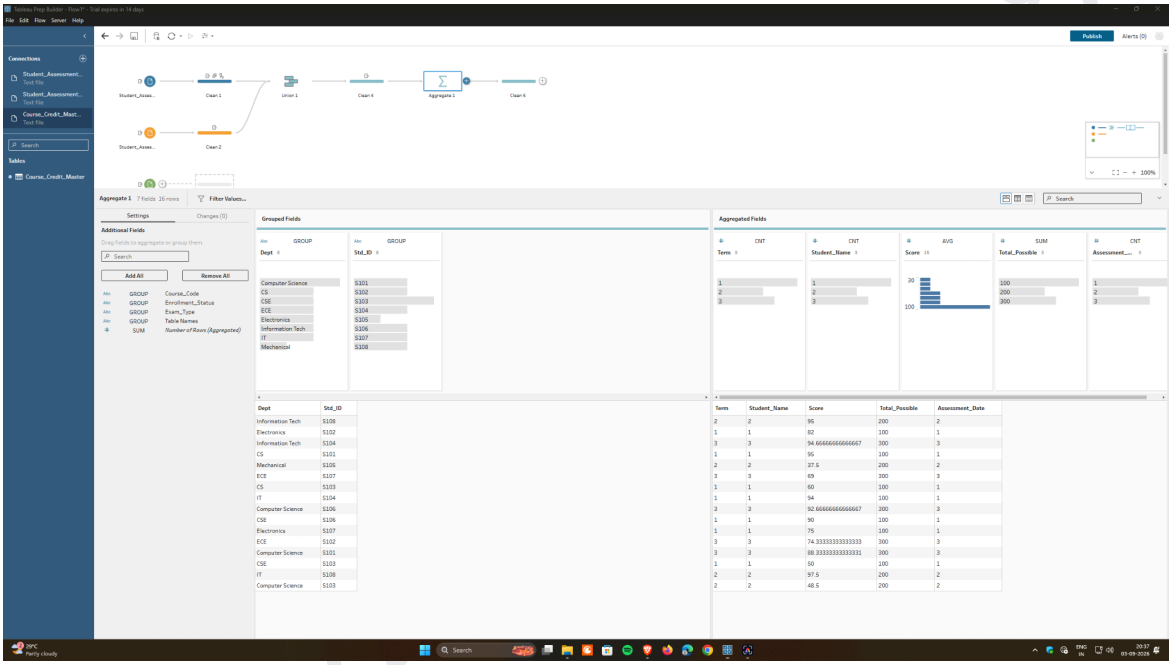
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Step 5: Rename the aggregated fields as follows: **Marks_Obtained (Sum) → Total_Marks_Obtained, Marks_Obtained (Average) → Average_Score, Max_Marks (Sum) → Total_Max_Marks, and Assessment_Type (Count) → Assessments_Taken**.

Step 6: Click + next to **Aggregate 1**, select **Clean Step** to create **Clean 4**, and verify that the **30 assessment rows** are reduced to about **15 summary records** (one row per student per term).

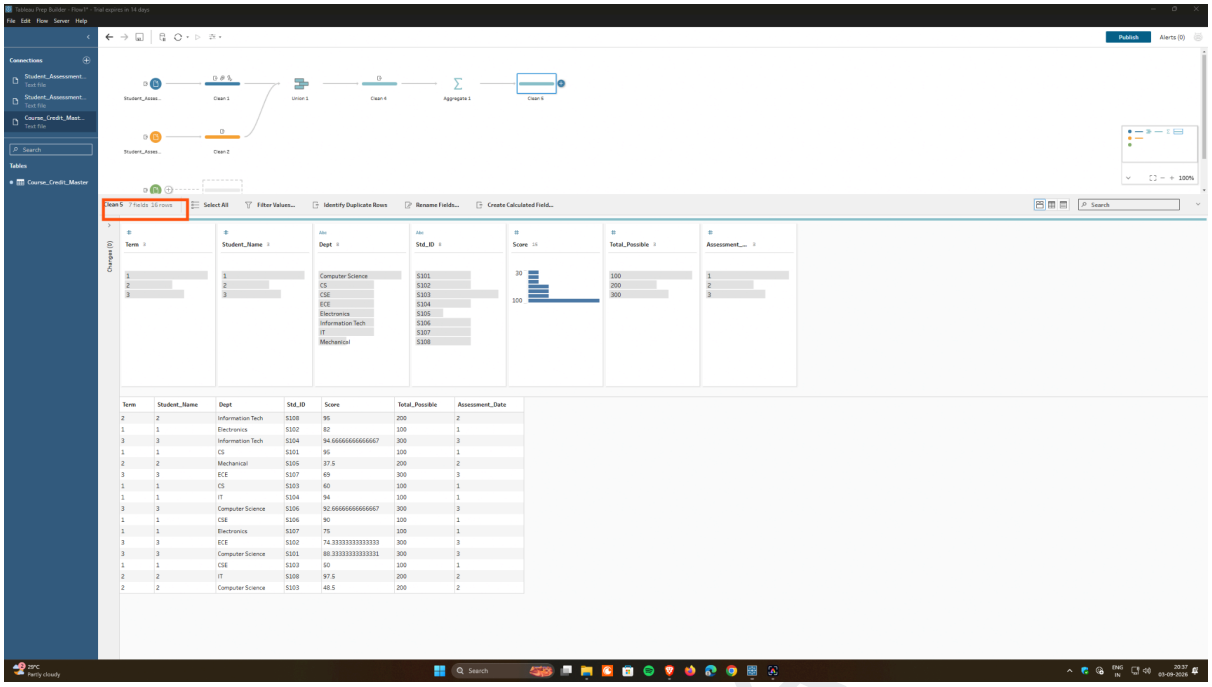


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F: Apply Filters and Transformations

Objective: Remove inactive/dropped records, compute weighted percentages, assign automated grading tiers, and derive academic standing indicators using calculated fields in Tableau Prep.

Not all raw records should enter downstream analytical dashboards (e.g., students who dropped out mid-term skew department averages). Transforming raw totals into percentage metrics and standardized letter grades directly inside the data prep pipeline ensures that business logic is applied once and remains consistent across all downstream reporting tools.

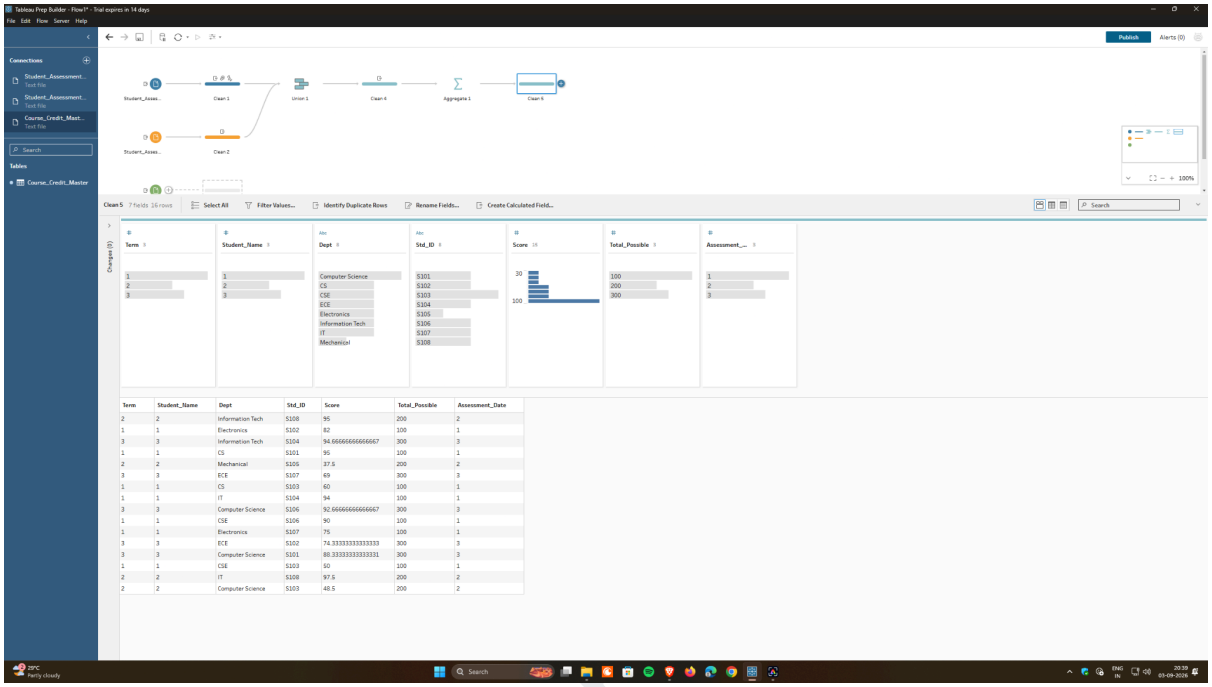
Step 1: Select **Clean 4** after **Aggregate 1** and verify the Profile Pane shows the aggregated fields: **Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken** (names may vary based on your Union).

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Term	Student_Name	Dept	Std_ID	Score	Total_Possible	Assessment_Date
2	2	Information Tech	S108	95	200	2
1	1	Electronics	S102	82	100	1
3	3	Information Tech	S104	94.66666666666667	300	3
1	1	CS	S101	96	100	1
2	2	Mechanical	S106	37.5	200	2
3	3	ECE	S107	69	300	3
1	1	CS	S103	60	100	1
1	1	IT	S104	94	100	1
3	3	Computer Science	S106	92.66666666666667	300	3
1	1	CSE	S106	90	100	1
1	1	Electronics	S107	75	100	1
3	3	ECE	S102	74.33333333333333	300	3
3	3	Computer Science	S101	88.33333333333331	300	3
1	1	CSE	S103	50	100	1
2	2	IT	S108	97.5	200	2
2	2	Computer Science	S103	48.5	200	2

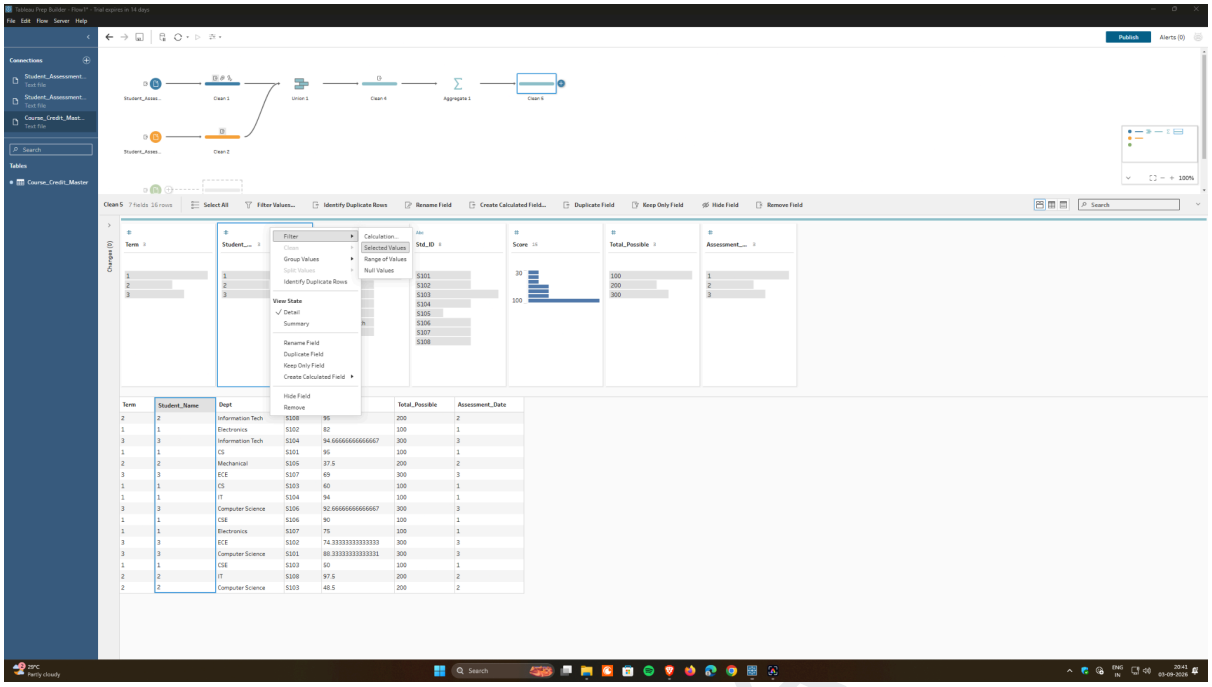
Step 2: If **Enrollment_Status** is available, filter it to keep only **Active** students by excluding **Dropped** records using **Exclude** or **Filter Values → Selected Values**.

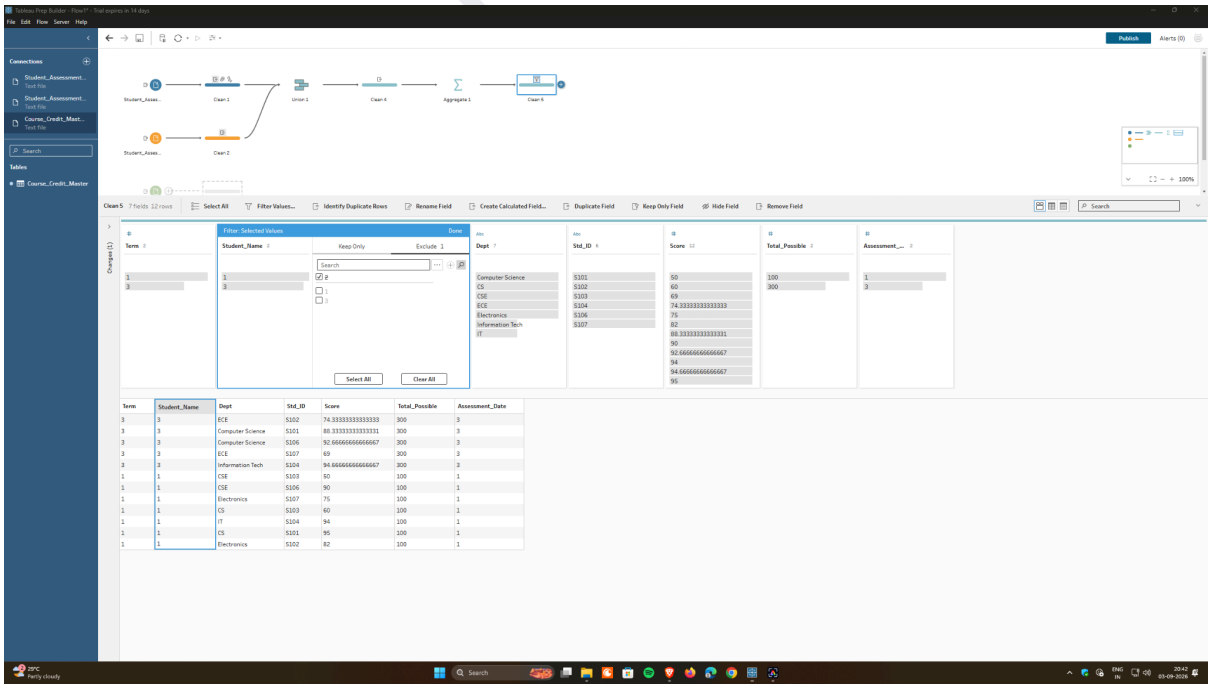
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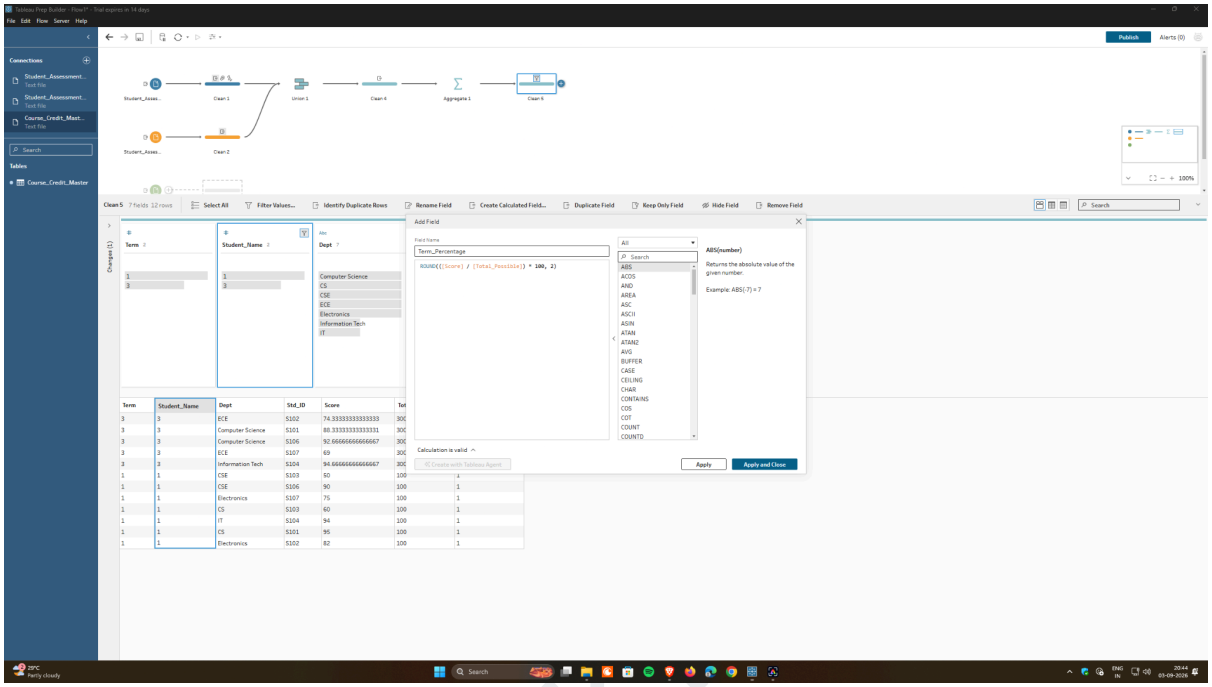


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Step 3: If Enrollment_Status is available, filter it to keep only Active students by excluding Dropped records using Exclude or Filter Values → Selected Values.



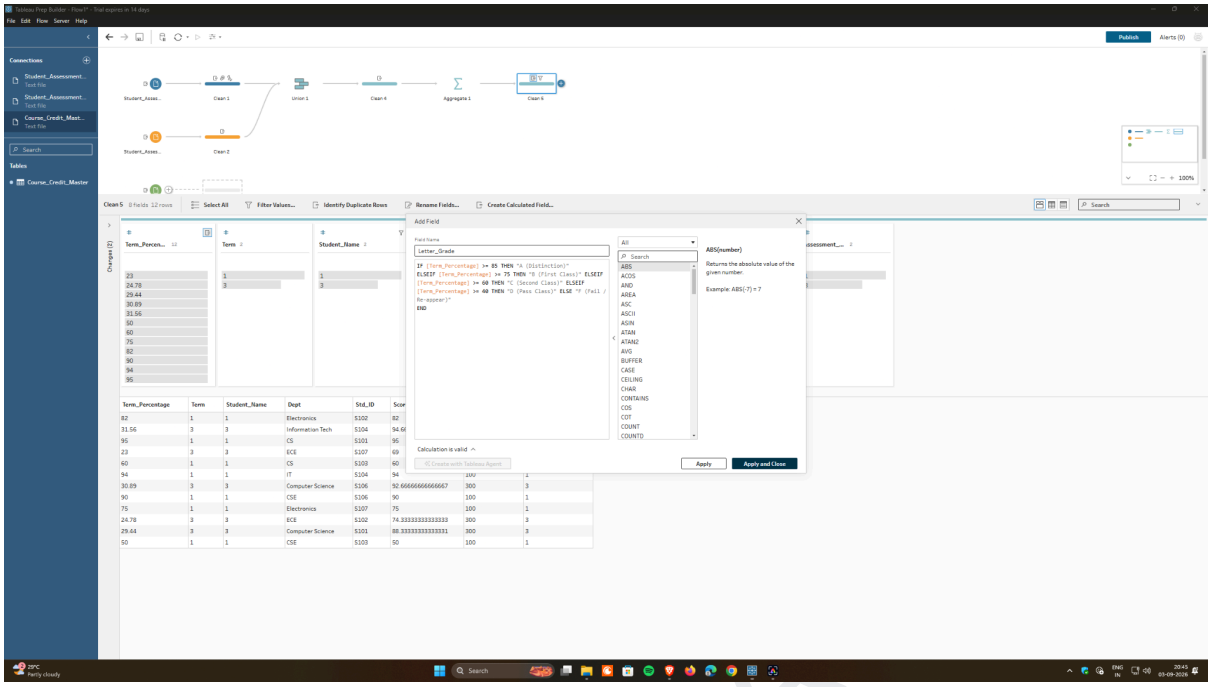
Step 4: Create a calculated field named **Letter_Grade** using the given **Term_Percentage** thresholds to classify students from **A (Distinction)** to **F (Fail / Re-appear)**, then click **Save**.

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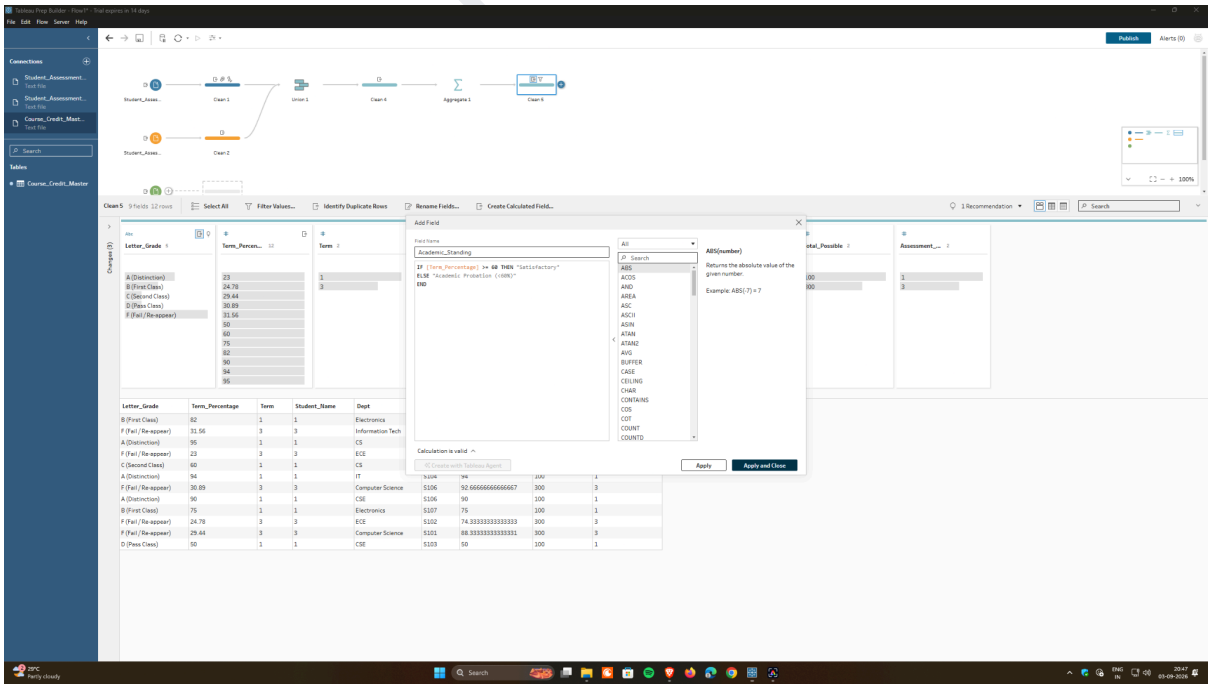
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Step 5: Create a calculated field named Academic_Standing using IF [Term_Percentage] >= 60 THEN "Satisfactory" ELSE "Academic Probation (<60%)" END.



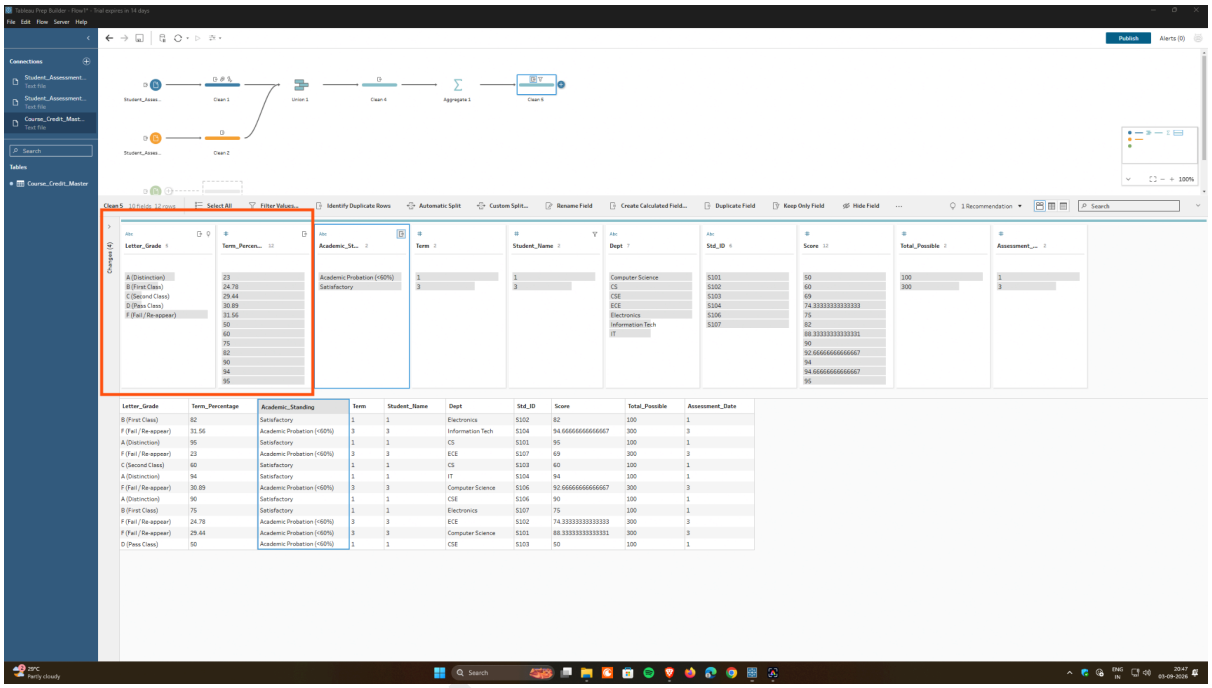
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Step 6: Verify Clean 4 to ensure Term_Percentage ranges from 0–100%, Letter_Grade shows categorical grade buckets, and Academic_Standing identifies students needing academic counseling.

.



Transformation Summary Table

Transformation	Type	Formula / Rule	Output Meaning
Filter Status	Row Exclusion	[Enrollment_Status] != 'Dropped'	Removes dropped students
Term_Percentage	Numeric Calculation	([Total_Marks_Obtained] / [Total_Max_Marks]) * 100	Normalized 0–100 scale
Letter_Grade	Conditional Dimension	IF [Term_Percentage] >= 85 THEN "A (Distinction)" ...	Standardized letter grade

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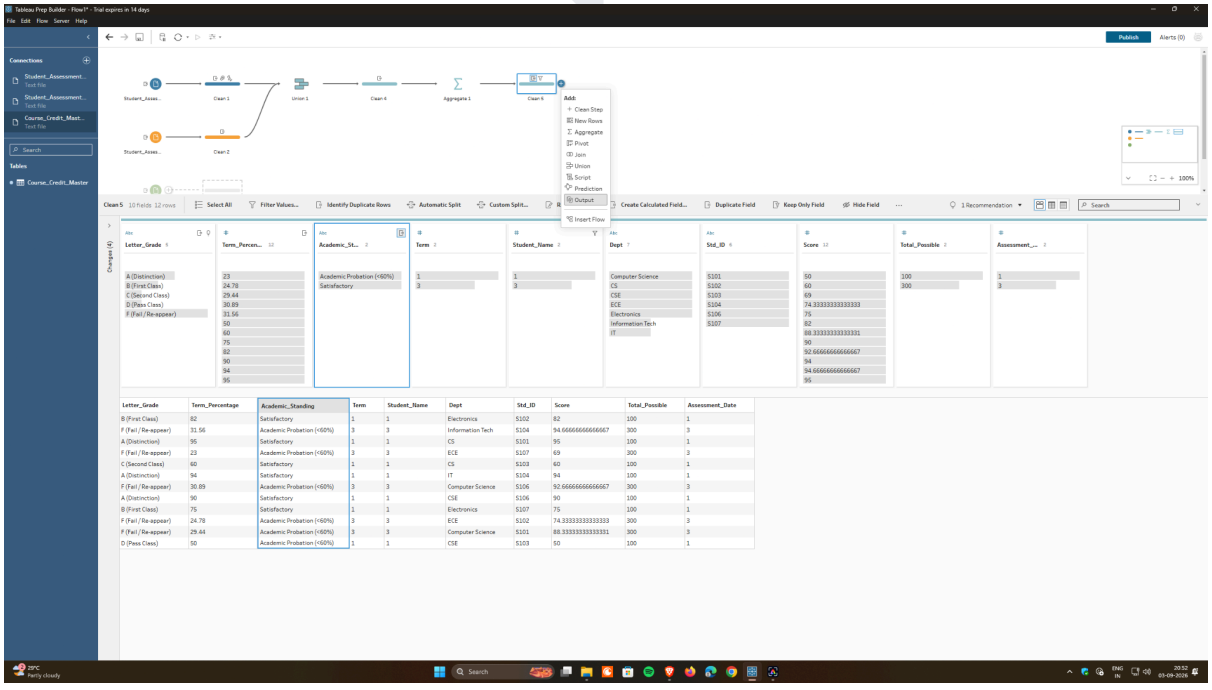
Academic_Standi ng	Conditional Flag	IF [Term_Percentage] >= 60 THEN "Satisfactory" ...	Compliance warning tag

G: Create and Automate Data Preparation Flows

Objective: Configure an Output Step in Tableau Prep Builder to publish the fully transformed pipeline to an extract file (.hyper or .csv), and learn how to automate or execute the flow on demand.

Data preparation workflows must be repeatable and production-ready. Configuring a dedicated output node ensures that whenever new assessment scores arrive, running the flow automatically executes all trimming, grouping, merging, filtering, and scoring calculations in seconds without any manual rework.

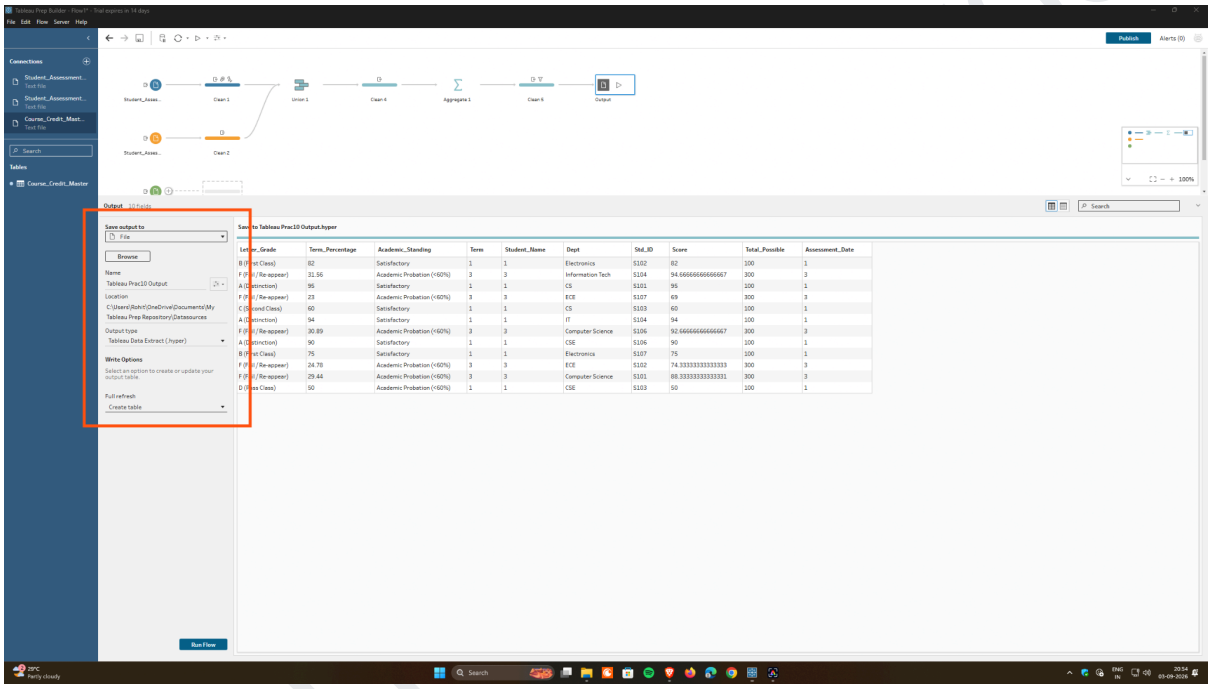
Step 1: Hover over **Clean 4**, click **+**, select **Output** (download/save icon), and verify the **Output** node is connected directly to **Clean 4**.



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Step 2: Select the **Output** node and configure **Save to File**, choose **.hyper** or **.csv**, set the desired **Location**, name the file **Student_Performance_Master_Cleaned**, and select **Create table** as the write option.

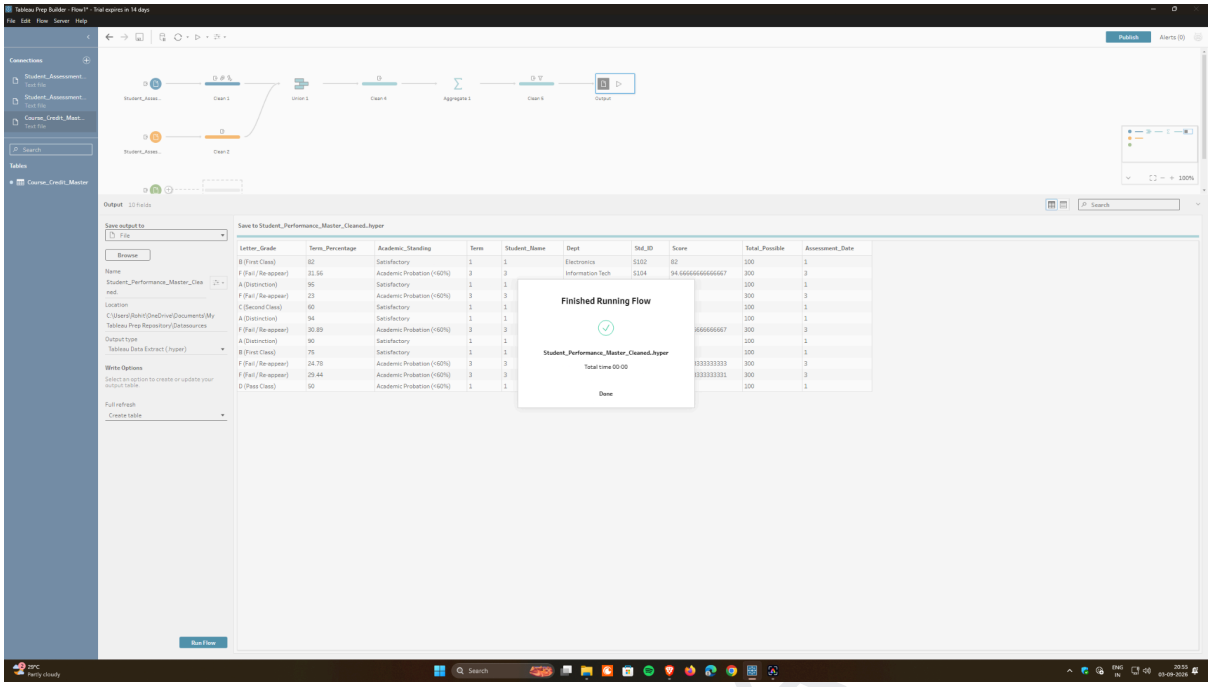


Step 3: Click **Run Flow** (blue button) or the ▸ icon on the Output node to process the entire pipeline, then verify the **"The flow ran successfully in X seconds."** notification appears.

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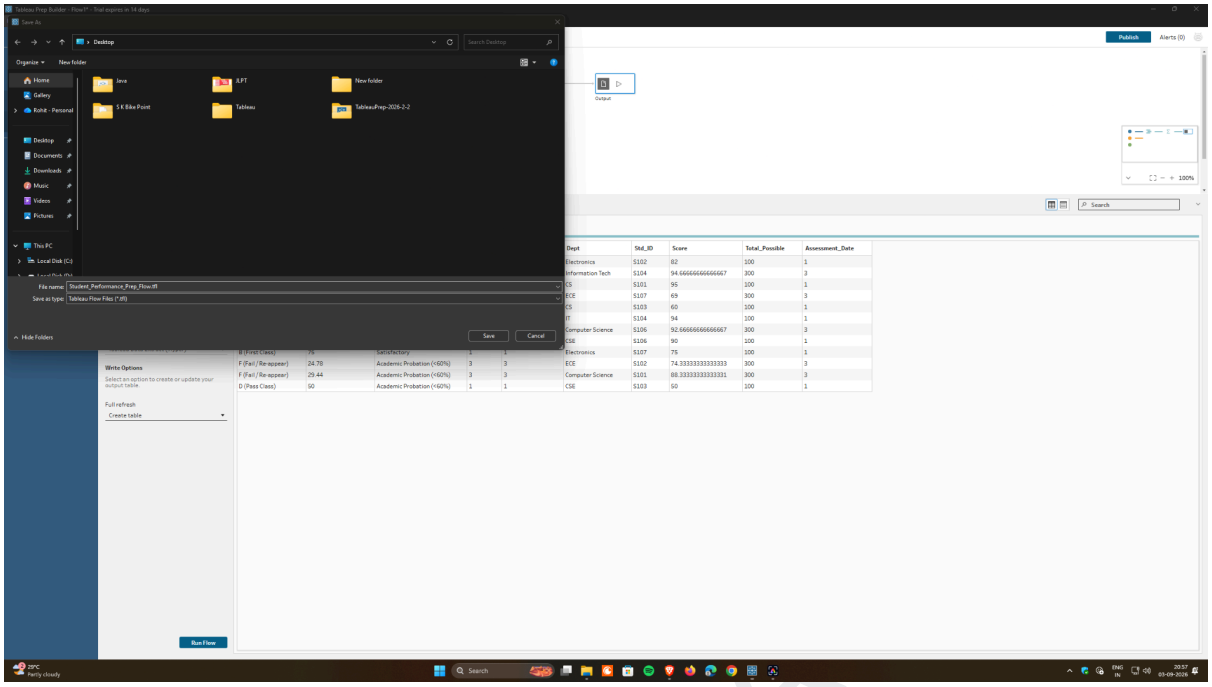
Step 4: Go to **File** → **Save As**, save the workflow as **Student_Performance_Prep_Flow.tfl**, and optionally automate it using **Tableau Prep CLI** or **Tableau Prep Conductor** with a scheduled refresh.

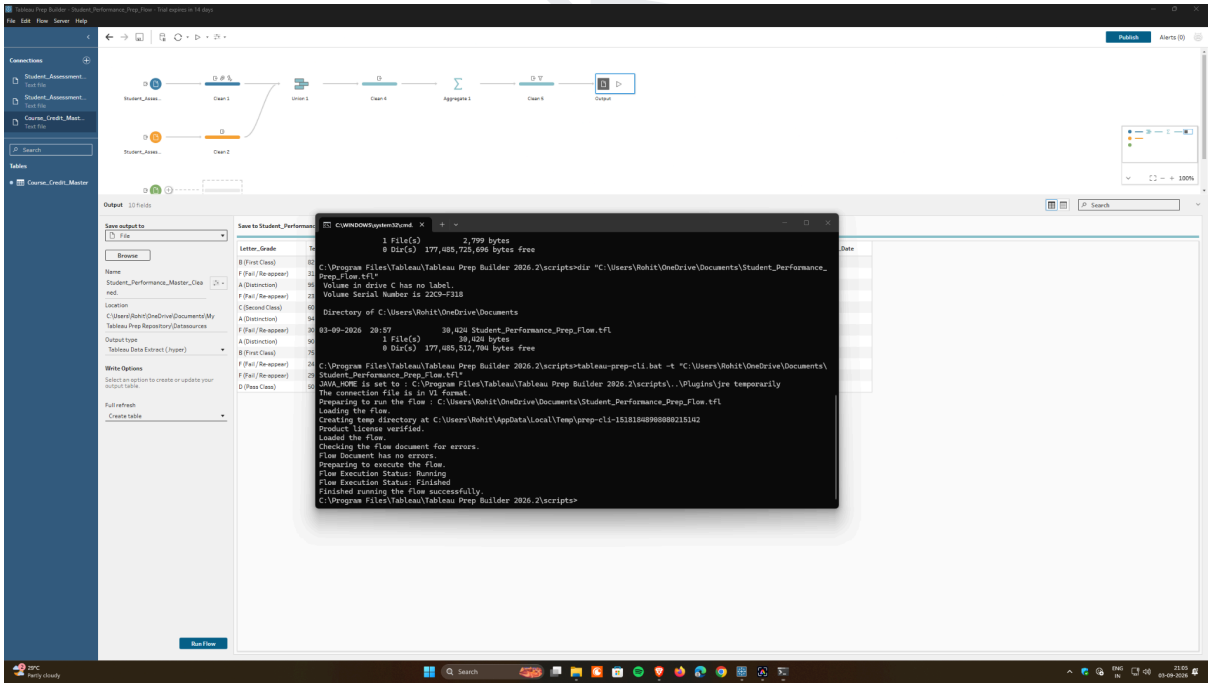
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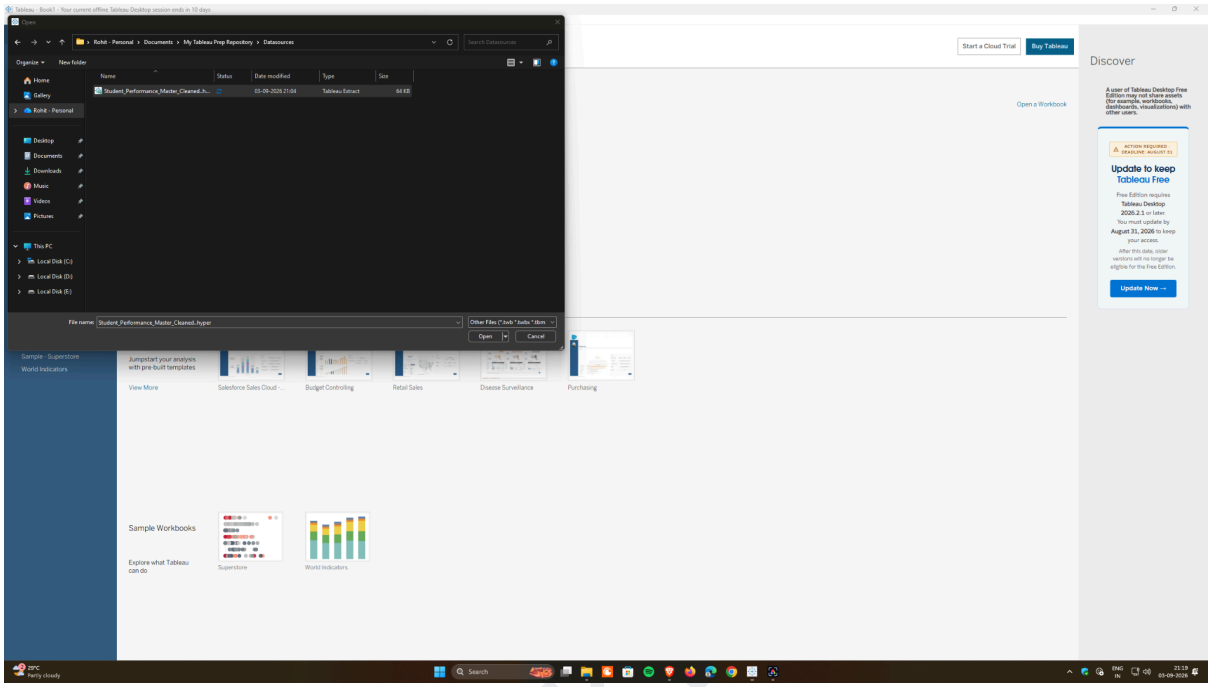
Step 5: Open **Tableau Desktop**, connect to **Student_Performance_Master_Cleaned.csv/.hyper**, then create the dashboard

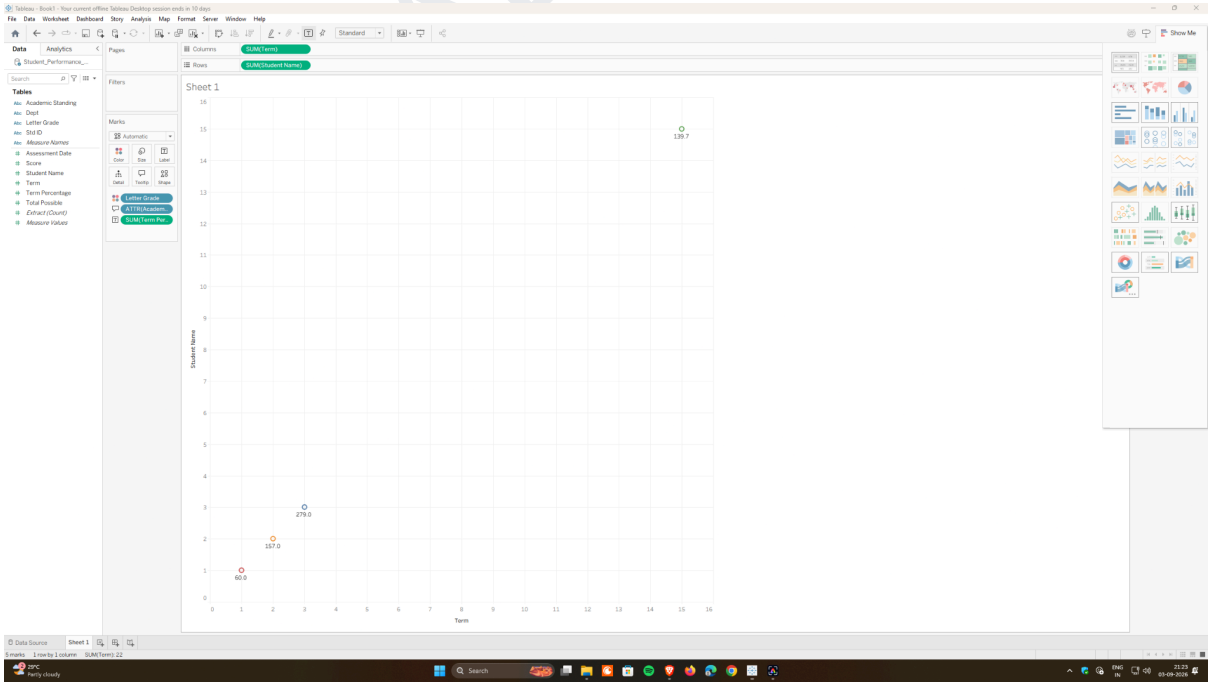
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by placing **Student_Name → Rows, Term → Columns, Term_Percentage → Text, Letter_Grade → Color, and Academic_Standing → Tooltip**.



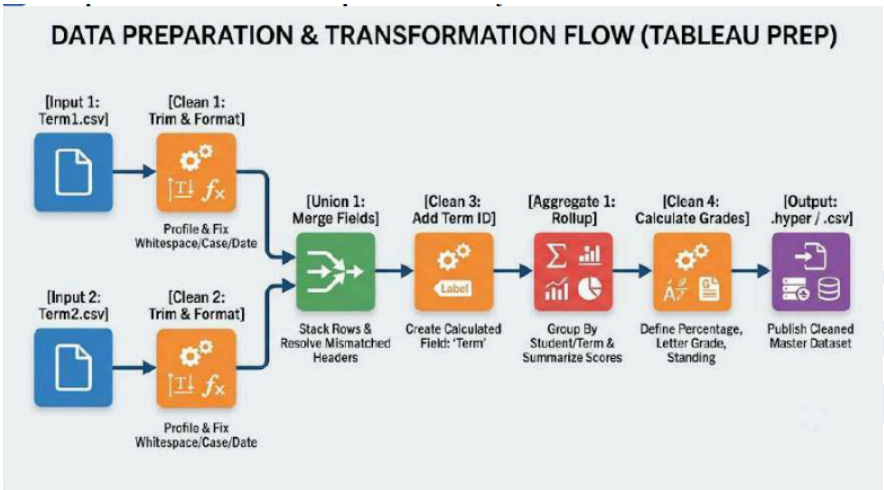


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Complete End-to-End Flow Pipeline Summary



Pipeline Stage	Step Type	Key Actions Performed	Output Result
Input Layer	Data Ingestion	Connect to Term1.csv, Term2.csv, and Course_Master.csv	Raw tables loaded on canvas
Cleaning Layer	Clean Steps 1 & 2	Trim whitespace, standardize dates, uppercase names	Structured singlesource tables
Standardization	Group Values	Consolidate aliases (CS, CSE, Comp Sci$\rightarrow$ Computer Science)	Clean department dimensions
Integration Layer	Union Step	Stack Term 1 & 2; drag-and-drop mismatched headers	Unified schema (30 assessment rows)
Aggregation Layer	Aggregate Step	Group by student & term; compute Sum(Marks) and Avg(Score)	14 summary records
Business Logic	Clean Step 4	Filter dropped students; calculate %, Letter_Grade, and Standing	Analytical scorecard metrics

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Export Layer	Output Step	Export to Student_Performance_Master_Cleaned .hyper / .csv	Production-ready dashboard source

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